Predicting the Likelihood of Type 2 Diabetes (T2D)

Team 3c

2026.02.08

CDS-492 | Spring 2026 | Dr. Slamani

Module 01 Participants:

Tyson Johnson
Nithila Neethi Devan
Alizeh Murtaza

Outline

- MD01 – First Team Meeting
  - Define roles
    - Team Lead: Tyson Johnson
    - Note taker/task scheduler: Nithila Neethi Devan
    - Assignment reviewer/submitter: Alizeh Murtaza
  - Discuss choices and ranking of the two project topics
    - Topic J : How can we use patient data to predict the likelihood of developing Type-2 Diabetes (T2D)?
    - Topic H: How can a hospital optimize the allocation of its resources to improve patient outcomes?
  - Submit assignment

Meeting Minutes

Date: 01/24/2026

Participants:

- Tyson Johnson
- Nithila Neethi Devan
- Alizeh Murtaza

Agenda:

- Discuss positions/roles within the team
- Discuss data source options
- Future Assignment Planning

Notes:

- Team Selected Topic J (T2D) as project focus
- Potential Data Source: South Korean Government Data Portal
  - Tyson has existing logic to translate data via Python to English.
- Literature Review Requirement: Each member should find 2-3 sources prior to next meeting; we can discuss strongest sources/what to include during the next meeting.
- Team has decided to continue with virtual meetings; next meeting is set for Jan 31st at 5pm (flexible)

Meeting Minutes - Tasks

Task No.	Task Name	Description	Owner	Date	Status
T01	Collect Potential Sources	Each team member will find 2-3 sources for the literature review for discussion during next meeting.	Tyson, Nithila, Alizeh	01/24/2026	Started
T02					
T03					
T04					
T05					
T06					
T07					

Note:

- Status: Not started, Started, Waiting on Name of person, Date Completed
- For more tasks, add rows to the table by right clicking on a cell in the last row and choose from the horizontal menu “Insert Rows Below”
- Keep the same table for the following week(s) and add more rows

Module 02 Participants:

Tyson Johnson
Nithila Neethi Devan
Alizeh Murtaza

MD02 – Revisions to work from previous module(s)

- Updated title from given question to: Predicting the likelihood of T2D (Slide 1)
- Updated Task Description and Status (Slide 9)
- Minor Formatting Changed (Slide 4, Slide 5, Slide 8, Slide 9)

Meeting Minutes

Date: 02/01/2026

Participants:

- Tyson Johnson
- Nithila Neethi Devan
- Alizeh Murtaza

Agenda:

- Review first submission
- Discuss slide formatting
- Discuss literature review articles
- Complete "Data Search" slide
- Update tasks
- Schedule next meeting

Notes:

- Discussed Potential Data Sources and Factors to consider when finding data.
- Ask Professor if our question can be slightly modified to accommodate non-time-series data if longitudinal data are not publicly available.

Meeting Minutes - Tasks

Task No.	Task Name	Description	Owner	Date	Status
T01	Collect Potential Sources	Each team member will find 1 reliable source and complete our literature review and synthesis matrix on.	Tyson, Nithila, Alizeh	01/31/2026	01/31/2026
T02	Research Data Options	Find Data that tracks a unique identifier information, data that would allow for time series methods to be used.	Tyson, Nithila, Alizeh	01/31/2026	Started
T03					
T04					
T05					
T06					
T07					

Note:

- Status: Not started, Started, Waiting on Name of person, Date Completed
- For more tasks, add rows to the table by right clicking on a cell in the last row and choose from the horizontal menu “Insert Rows Below”
- Keep the same table for the following modules(s) and add more rows

MD02 – Literature Review | Paper 1

- Paper title: The diabetes risk score: A practical tool to predict T2D risk
- Paper citation:
  - Lindstrom, J., & Tuomilehto, J. (2003). The Diabetes Risk Score: A practical tool to predict T2D risk. *Diabetes Care*, *26*(3), 725–731. https://doi.org/10.2337/diacare.26.3.725
- Summary of what was investigated:
  - Study investigated whether T2D risk can be predicted using noninvasive patient data without laboratory tests
- Results:
  - Identified key predictive variables including age, BMI, waist circumference, history of hypertension treatment, history of elevated blood glucose, physical activity, and diet.
  - The resulting Diabetes Risk Score demonstrated strong predictive performance (AUC ≈ 0.85–0.87), with scores of 9 or higher indicating substantially increased risk.
- What processing algorithms did each paper use:
  - Multivariate logistic regression, Regression coefficients converted to categorical point-based risk scoring, ROC curve analysis
- What you liked about the paper:
  - Model relies on accessible, real-world patient data, making it practical for large-scale screening.
- What additions you can provide to the topic:
  - Include additional patient data such as family history, socioeconomic factors, and race/ethnicity

MD02 – Literature Review | Paper 2

- Paper title:
  - AUSDRISK: an Australian T2D Risk Assessment Tool based on demographic, lifestyle and simple anthropometric measures
- Paper citation:
  - Chen, L., Magliano, D. J., Balkau, B., Colagiuri, S., Zimmet, P. Z., Tonkin, A. M., Mitchell, P., Phillips, P. J., & Shaw, J. E. (2010). AUSDRISK: an Australian T2D Risk Assessment Tool based on demographic, lifestyle and simple anthropometric measures. *Medical Journal of Australia*, *192*(4), 197–202. https://doi.org/10.5694/j.1326-5377.2010.tb03478.x
- Summary of what was investigated:
  - The development and design of AUSDRISK.
- Results:
  - AROC: 0.78 (95% CI, 0.76–0.81), HL Chi-Squared: 4.1 (P = 0.85)
  - Sensitivity: 0.74, Specificity: 0.677, PPV: 0.127
- What processing algorithms did each paper use:
  - Multiple linear regression, scoring system relying on beta scores. AROC and chi-squared for validation.
- What you liked about the paper:
  - Explained a lot of their decision-making process on things like certain features' inclusion/exclusion.
- What additions you can provide to the topic:
  - By using a different dataset and similar methodology, we can strive to reproduce similar results.

MD02 – Literature Review | Paper 3

- Paper title:
  - Opportunistic screening of T2D with deep metric learning using electronic health records
- Paper citation:
  - Jin, Q., Zhang, H., Szczerbinski, L., Zhu, J., Gerych, W., Xu, X., Wang, K., Hsu, S., Mandla, R., Deutsch, A. J., Manning, A., Mercader, J. M., Hartvigsen, T., Udler, M. S., & Ghassemi, M. (2025). Opportunistic screening of T2D with deep metric learning using electronic health records. *Scientific Reports*, *15*(1). https://doi.org/10.1038/s41598-025-25759-x
- Summary of what was investigated:
  - Researchers studied whether existing patient health records could be used to predict who will develop T2D in the future using ML models.
- Results:
  - The ML model predicted Tupe 2 diabetes more accurately than traditional statistical methods. It was able to identify high risk patients earlier by detecting the various patterns that can lead to patients developing diabetes.
- What processing algorithms did each paper use:
  - Deep Metric Learning trained on Electronic Health Records
- What you liked about the paper:
  - It combines both predicting diabetes onset and identifying patient subtypes in one model, this allows for more personalized pattern detecting, and therefore interventions
- What additions you can provide to the topic:
  - The incorporation of lifestyle or social factors to improve prediction accuracy.
  - Use of explainable AI to make predictions and pattern detection process more transparent and interpretable.

MD02 – Synthesis Matrix | Paper 1

Sources	Research Question/Hypothesis	Methods	Findings	Strengths	Limitations
Lindström, J., & Tuomilehto, J. (2003). The diabetes risk score: A practical tool to predict T2D risk. Diabetes Care, 26(3), 725–731.	The paper addresses whether T2D risk can be accurately predicted using noninvasive, easily collected patient data. The authors hypothesize that a simple risk score, based on demographic, anthropometric, behavioral, and medical history variables, can identify individuals at high risk for developing drug-treated T2D.	Population-based cohort study using Finnish FINRISK surveys (1987 and 1992). Participants: adults aged 35–64 without antidiabetic drug treatment at baseline. Follow-up: 5–10 years using national drug registry data. Multivariate logistic regression used to identify significant predictors.	A risk score ≥9 effectively identified individuals at high risk for future drug-treated T2D. The model demonstrated strong predictive performance (AUC ≈ 0.85–0.87). The score also performed well in identifying undiagnosed diabetes in cross-sectional analyses.	Uses noninvasive, low-cost, easily accessible patient data. Strong validation across independent cohorts. Practical and scalable for population-level screening.	Study population limited to Finnish adults, reducing generalizability. Outcome defined as drug-treated diabetes, potentially missing earlier or diet-controlled cases. Use of categorical scoring may reduce predictive precision.

MD02 – Synthesis Matrix | Paper 2

Sources	Research Question/Hypothesis	Methods	Findings	Strengths	Limitations
Chen, L., Magliano, D. J., Balkau, B., Colagiuri, S., Zimmet, P. Z., Tonkin, A. M., Mitchell, P., Phillips, P. J., & Shaw, J. E. (2010). AUSDRISK: an Australian T2D Risk Assessment Tool based on demographic, lifestyle and simple anthropometric measures. *Medical Journal of Australia*, *192*(4), 197–202. https://doi.org/10.5694/j.1326-5377.2010.tb03478.x	"To develop and validate a diabetes risk assessment tool for Australia based on demographic, lifestyle and simple anthropometric measures." The authors explain their development of AUSDRISK.	9 risk factors categorized Pearson chi-squared test Prediction models made using multiple linear regression. Risk factors: age, sex, ethnicity, parental history of diabetes, history of high blood glucose level, use of antihypertensive medications, smoking, physical inactivity, waist circumference	AusDiab data: 1999-2000, 11,247 adults, Ages 25+, Australian 5-year follow-up: 2004-2005, 6,537/10,788 surviving returned Used 6,060 participants after removing 246 unclassifiable and 229 baseline diabetic (from AusDiab). AROC: 0.78 (95% CI, 0.76–0.81) HL Chi-Squared: 4.1 (P = 0.85) Sensitivity: 0.74 Specificity: 0.677 PPV: 0.127	The data used included a wide variety of features measured that other models may not have used, such as the use of certain medications and parental history of diabetes. The source provides a working example of a model used for predicting T2D. Can be accessed as a free "online interactive risk assessment tool." http://health.gov.au/internet/main/publishing.nsf/Content/diabetesriskassessmenttool	Acknowledges that data limitations would likely result in overly raised risks in ages under 25. May have also benefitted from a more ethnically diverse pool of participants for broad use.

MD02 – Synthesis Matrix | Paper 3

Sources	Research Question/Hypothesis	Methods	Findings	Strengths	Limitations
Jin, Q., Zhang, H., Szczerbinski, L., Zhu, J., Gerych, W., Xu, X., Wang, K., Hsu, S., Mandla, R., Deutsch, A. J., Manning, A., Mercader, J. M., Hartvigsen, T., Udler, M. S., & Ghassemi, M. (2025). Opportunistic screening of T2D with deep metric learning using electronic health records. *Scientific Reports*, *15*(1). https://doi.org/10.1038/s41598-025-25759-x	Can routine electronic health records be used to predict the future onset of T2D and identify patient subtypes?	Trained a deep metric learning model on longitudinal EHR data to predict future T2D onset and learn patient patterns Used these learned patterns to cluster patients into distinct subtypes Compared model performance with logistic regression and other baseline methods	The DML model outperformed traditional prediction methods for future T2D onset prediction and revealed three distinct subtypes of patients with different clinical characteristics and responses to treatment. AUROC of *0.969* Sub types : Green (healthier, more responsive), Yellow (intermediate), Red (high comorbidity burden)	Combines disease prediction and patient subtyping within a single model Utilizes large, diverse EHR datasets Enables practical screening using existing clinical data without additional testing	Does not include lifestyle or social determinants of health Limited interpretability due to the black-box nature of deep learning

MD02 – Data Search

- Data sources
  - https://www.data.go.kr/en/data/15007122/fileData.do
- Relevance of the data to the project
  - Dataset contains features that is publicly available.
  - Dataset contains yearly health exam results of South Korean patients.
- Possible challenges with the data
  - Data is specific to the Korean demographic
  - Unsure if there is a way to incorporate time series methods as we are still identifying data with unique identifier information (subscriber serial number)

Module 03 Participants:

Tyson Johnson
Nithila Neethi Devan
Alizeh Murtaza
Tariq Abdulhak

MD03 – Revisions to work from previous module(s)

- Below is a summary of changes made on previous modules and slide numbers:
  - File name correction
  - Added Tariq Abdulhak to title slide (Slide 1)
  - “Type-2 Diabetes” to “T2D” abbreviation changes

Meeting Minutes

Date: 02/05/2026

Participants:

- Tyson Johnson
- Nithila Neethi Devan
- Alizeh Murtaza
- Tariq Abdulhak

Agenda:

- Folder Structure
- Workflow Diagram
- Business Understanding
- Data Search and Description

Notes:

- Allocated Tasks per person between Workflow Diagram, Business Understanding, and Data Search post discussion
- Discussed a change in our business problem, shift the focus off of longitudinal data
- Finished on Business Understanding together based on up

Meeting Minutes - Tasks

Task No.	Task Name	Description	Owner	Date	Status
T01	Collect Potential Sources	Each team member will find 1 reliable source and complete our literature review and synthesis matrix on.	Tyson, Nithila, Alizeh	01/31/2026	02/05/2026
T02	Research Data Options	Find Data that tracks a unique identifier information, data that would allow for time series methods to be used.	Tyson, Nithila, Alizeh	02/05/2026	02/05/2026
T03	Workflow Diagram	Creating the workflow diagram through Xmind	Nithila	02/05/2026	Started
T04	Black Box Diagram	Creating the back box diagram through canva	Alizeh	02/05/2026	Started
T05	Class Folder Structure	Created and Shared Class Folder	Alizeh	02/05/2026	Started
T06	Fishbone Diagram	Creating the black box diagram through canva	Alizeh	02/05/2026	Started
T07	Business Understanding	Completing the business understanding slide	Tariq, Alizeh, Tyson, Nithila	02/05/2026	02/05/2026
T08	Data Search	Completing the Data Source Slides	Tyson	02/05/2026	Started
T09	Analytic Problem	Completing the analytic problem slide	Tariq	02/05/2026	Started

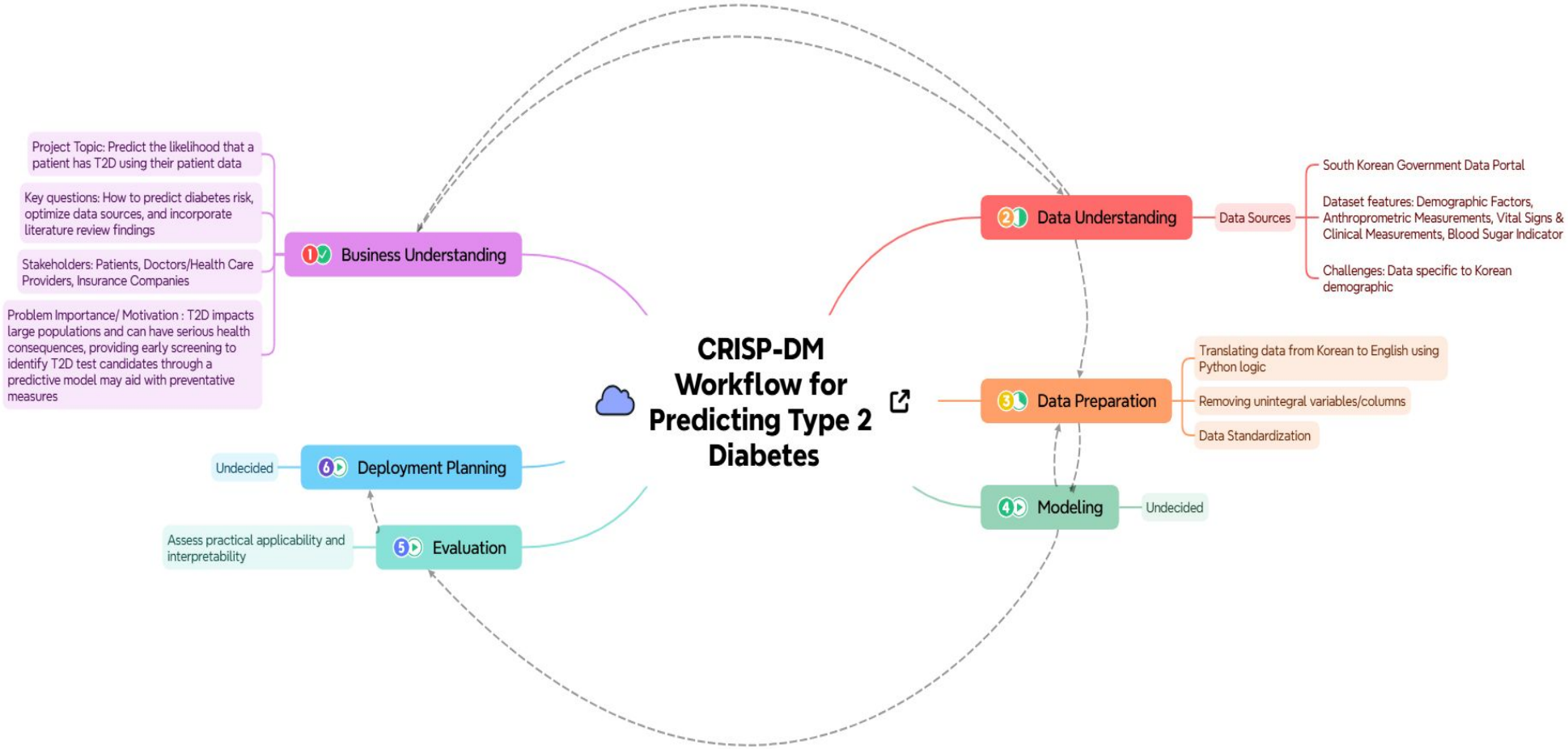
MD03 – Class Folder

- Class folder structure:

```
CDS-492
 ├──── Code
 ├──── Data
|   └──── Set_01
 └──── Project
     └──── Assignments
         ├────
WK01_Team_3c_CDS-492_Project_SubmissionDate_2026-1.24.2026.pdf
         └────
WK02_Team_3c_CDS-492_Project_SubmissionDate_2026-2.01.2026.pdf
```

MD03 – Workflow Diagram

- Workflow Diagram as of 2026.02.05



MD03 – Business Understanding

- Business problem and question(s)
  - Predict the likelihood that a patient has T2D using their patient data
- Problem importance
  - T2D is a significant problem that impacts the general population with serious health consequences
  - Diabetes is the 7th leading cause of death in the US (https://www.cdc.gov/nchs/fastats/diabetes.htm)
- Stakeholders
  - Patients
  - Doctors/Health care providers
  - Insurance companies
- Motivation
  - Provide early screening to identify T2D test candidates
- Suggested title for the project
  - Screening T2D Using Patient Data
- Possible challenges with the problem topic
  - May not be able to create a predictive model that accurately diagnoses patients with T2D using Korean data used in this project

MD03 – Analytic Problem

Analytic Problem:

- Use patient data to predict whether a person is likely to have T2D
- The goal is to help identify people who may need early screening

Data:

- Korean patient health dataset
- Includes :
  - age and gender
  - height, weight, and waist size
  - blood pressure values
  - blood sugar and cholesterol levels

Goal:

- Predict diabetes risk based on patient health measurements
- Support early detection and prevention

MD03 – Analytic Problem (cont.)

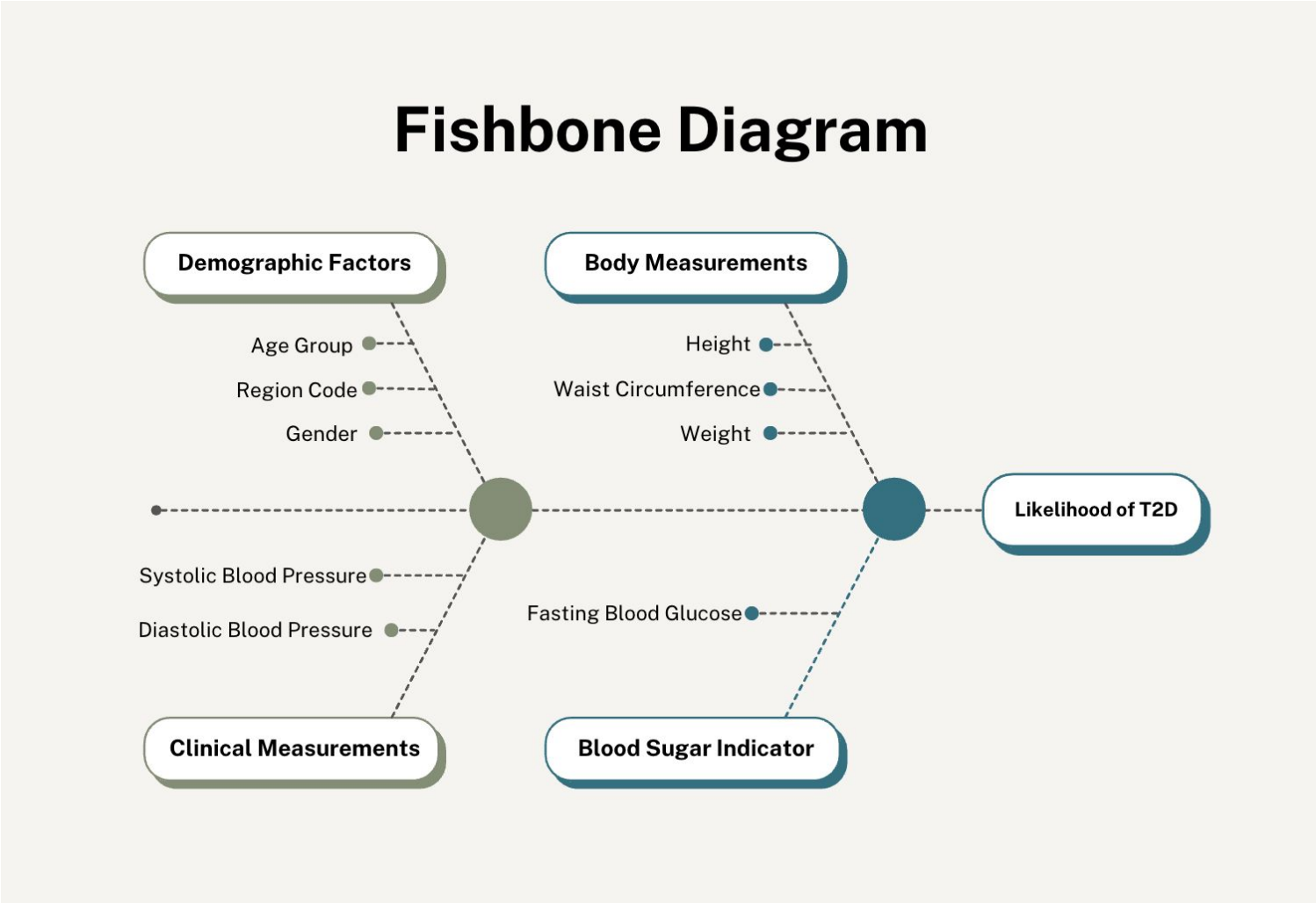
Assumptions and Limitations:

- The data is accurate and properly recorded
- Results may not apply to all populations

Important factors and categories:

- Demographics (age, gender)
- Body measurements (height, weight, waist size)
- Blood pressure
- Blood sugar levels

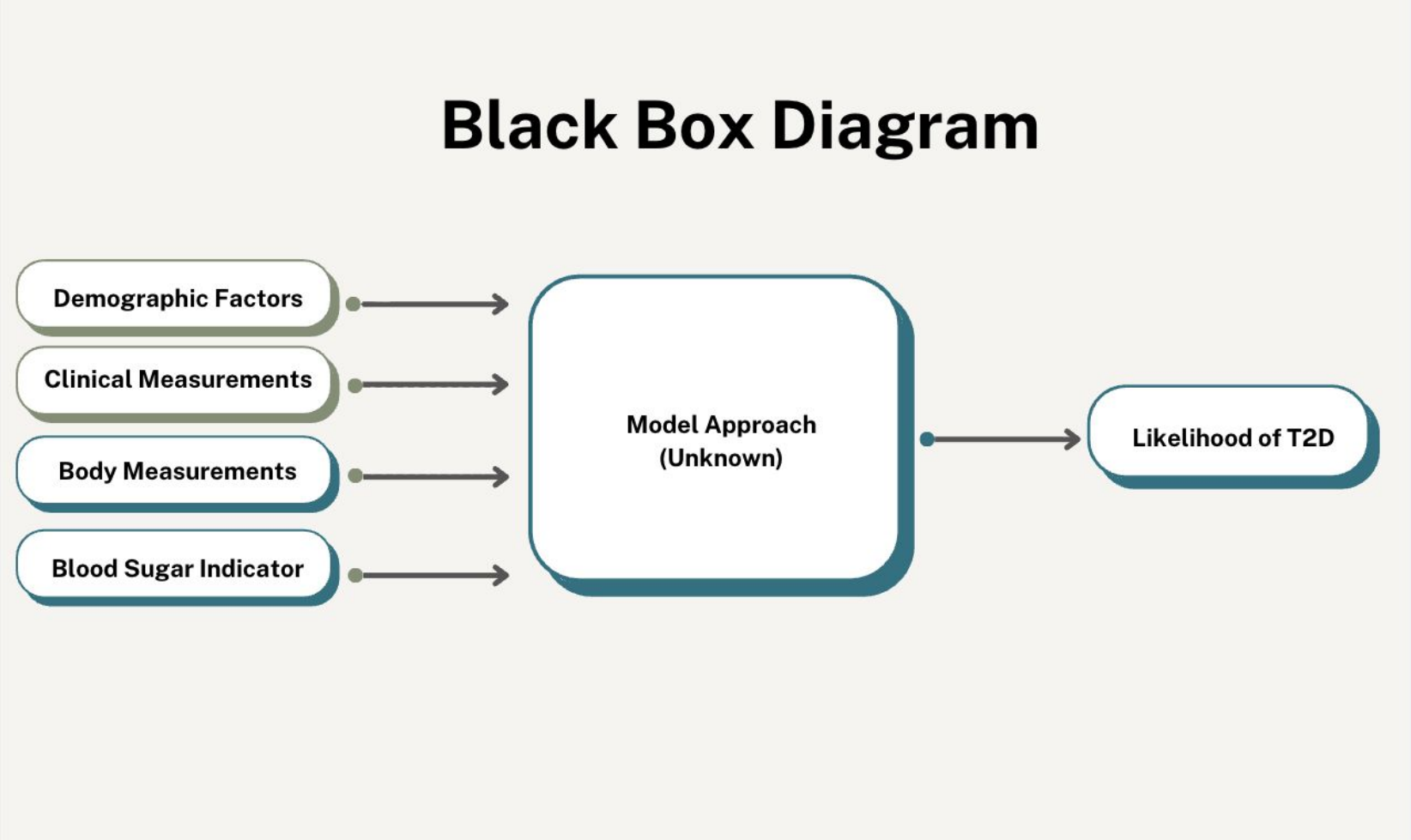
MD03 – Analytic Problem: Fishbone Diagram



MD03 – Analytic Problem: Black Box Diagram

- Inputs
  - Demographic Factors
  - Clinical Measurements
  - Body Measurements
  - Blood Sugar Indicator
- Outputs
  - Likelihood of T2D
- Modeling approach if known
  - Unknown

MD03 – Analytic Problem: Black Box Diagram



MD03 – Data Search

- Data sources
  - https://www.data.go.kr/en/data/15007122/fileData.do
  - Publicly provided by South Korea's National Health Insurance Service (NHIS)

- Relevance of the data to the project
  - Patient data of 1 million randomly selected patient exams throughout year.
  - Patient data from 2002-2024 (updated annually) available.
  - Vast majority of exams include resting blood glucose measurements.

- Possible challenges with the data
  - Some variables are rounded.
  - Data and manual recorded in Korean, requiring accurate translations.
  - Lack of identifying information prevents tracking patient over time.

MD03 – Data Description

- Entries: 1,000,000
  - Represents an insurance-covered patient aged 20+.
  - Reduced to 994,153 after removing entries missing target variable (fasting blood glucose).
- Features: 34
  - Some features are completely missing or irrelevant.
  - Basic patient details, health exam results, questionnaire information.
- CSV filetype (cp949 encoding)
- Column names are self-descriptive, but manual provides detailed explanations.
- Varying sparsity because not all patients received all tests.
  - 334,265 entries have values for all relevant variables.
- All data in this set is from 2024.

MD03 – Data Description (cont.)

Feature	Minimum	Maximum	Mean	Median	St. Deviation
age_group_code*	5	18	10.64	11	3.05
height**	130	195	162.78	165	9.34
weight**	30	135	64.60	65	14.03
waist_circumference	6.8	210	81.55	81.1	10.94
systolic_blood_pressure	60	271	122.69	121	14.59
diastolic_blood_pressure	11	182	75.22	75	10.36
fasting_blood_sugar	17	991	100.98	96	23.00
total_cholesteral	25	2204	196.79	195	43.42
triglycerides	1	4622	129.32	105	99.46
hdl_cholesterol	1	908	57.09	55	15.48
ldl_cholesterol	0	2174	114.47	113	39.31
hemoglobin	0.1	25	14.22	14.2	1.53
serum_creatinine	0.06	87.2	0.84	0.8	0.37

*Grouped as 5 years (ex. 20-24). **Rounded down to nearest 5.

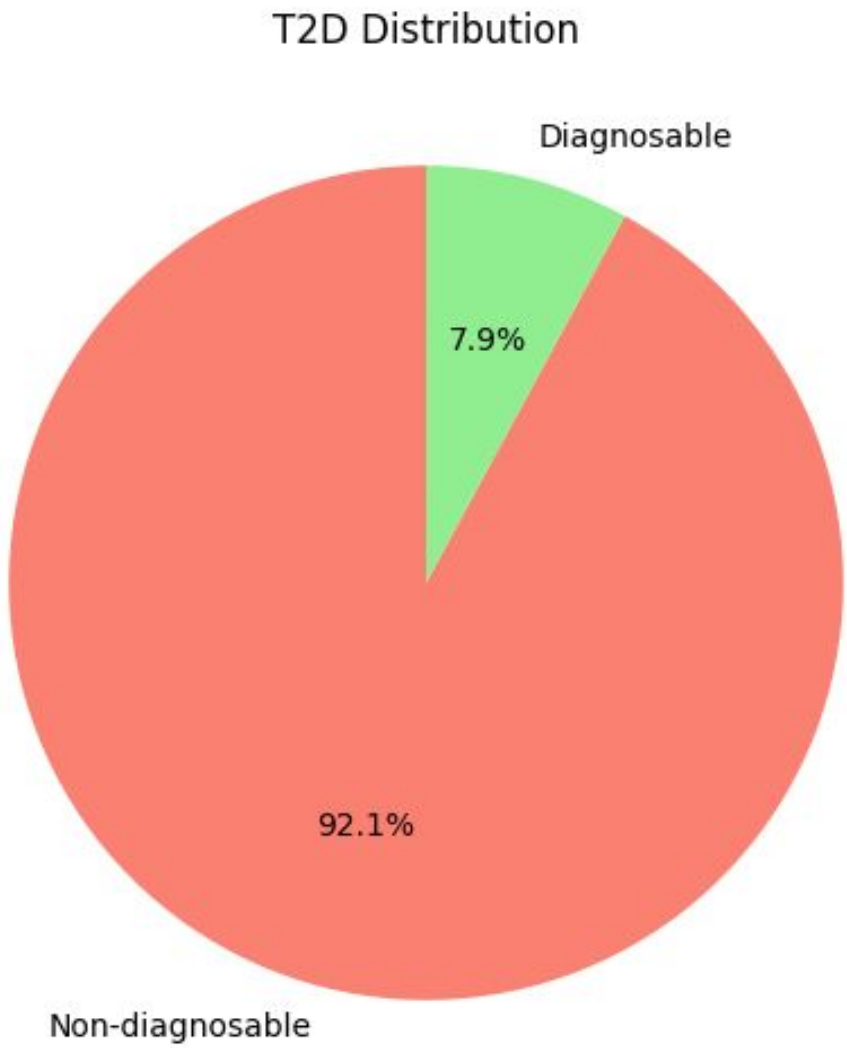
MD03 – Data Description (cont.)

Feature	Minimum	Maximum	Mean	Median	St. Deviation
serum_got_ast	1	7540	26.97	23	22.644
serum_gpt_alt	1	5310	26.53	21	25.52
gamma_gtp	1	9999	34.88	22	58.61

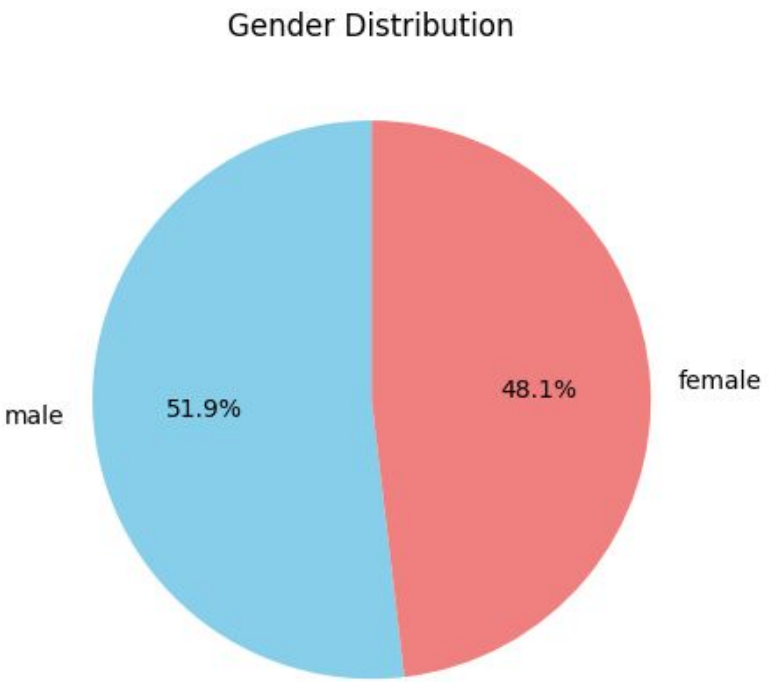
Non-numeric features recorded:

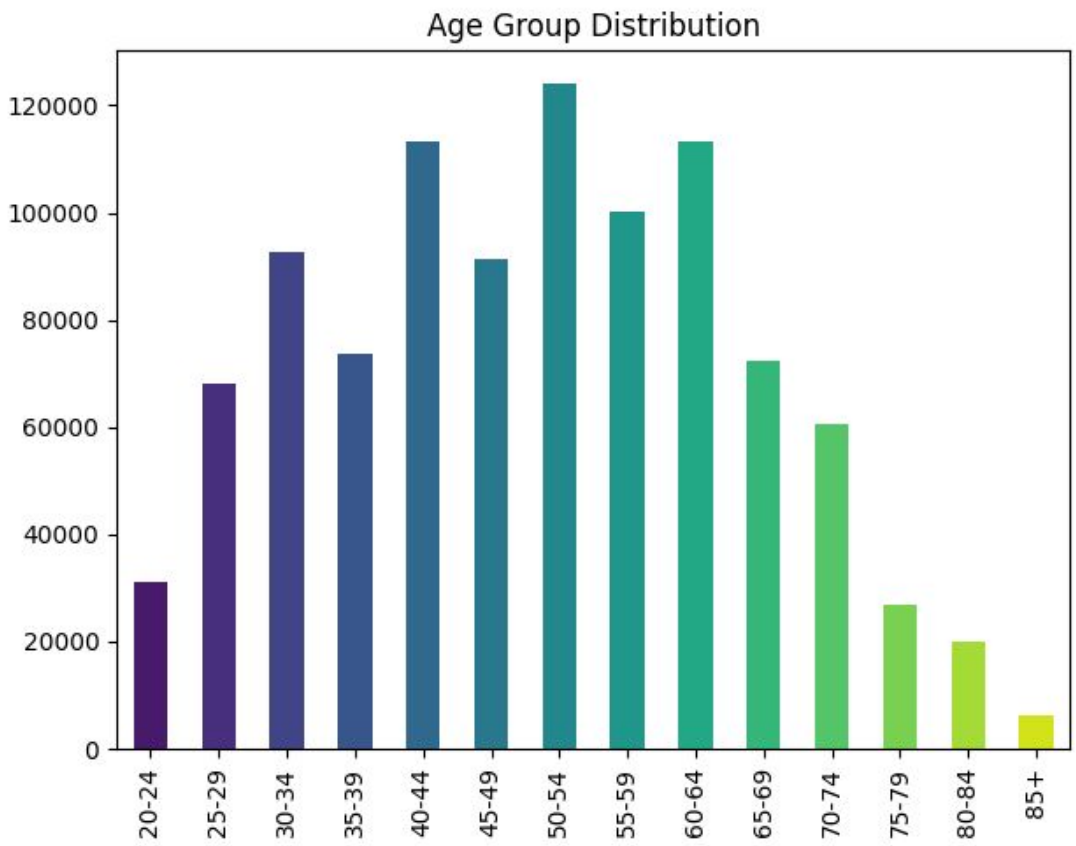
- gender_code (bool)
- vision_left/vision_right (has exception values)
- hearing_left/hearing_right (bool)
- urine_protein (categorical)
- smoking_status (categorical)
- alcohol_consumption (bool)
- oral_exam_status (bool)
- dental_caries_presence (bool)
- plaque_presence (categorical)

MD03 – Data Visualizations

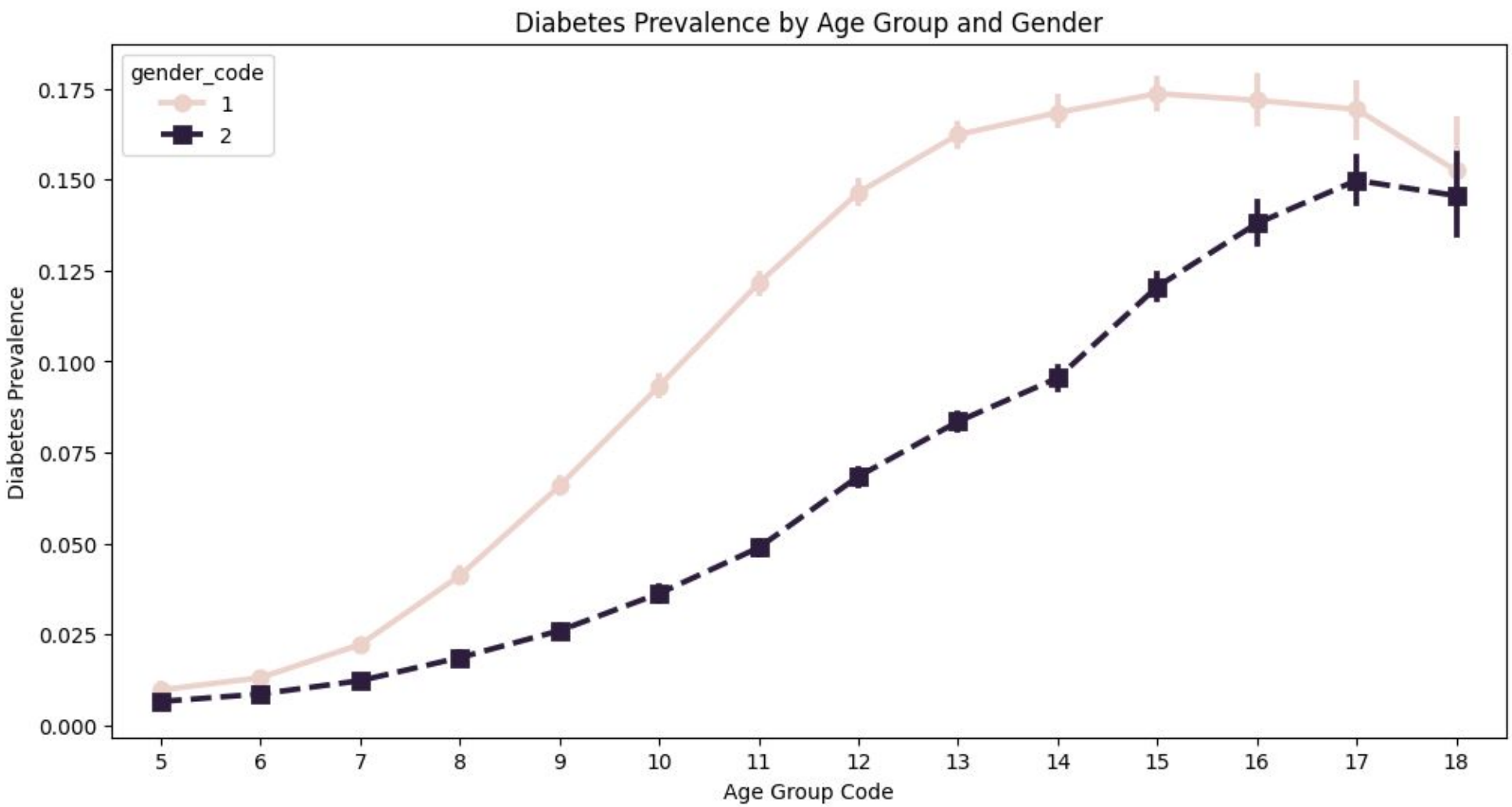


MD03 – Data Visualizations (cont.)





MD03 – Data Visualizations (cont.)



Module 04 Participants:

Tyson Johnson
Nithila Neethi Devan
Alizeh Murtaza
Tariq Abdulhak

MD04 – Revisions to work from previous module(s)

- Below is a summary of changes made to previous modules and their corresponding slide numbers:
  - Formatting Changes: font and bullet standardization
- Class folder was updated for this week
  - Link to online class folder: CDS-492

Meeting Minutes

Date: 02/12/2026

Participants:

- Tyson Johnson
- Nithila Neethi Devan
- Alizeh Murtaza
- Tariq Abdulhak

Agenda:

- Discussing feedback
- Reviewing CRISP-DM
- Assigning tasks
- Scheduling meetings

Notes:

- Meeting @ 3:30 Monday 2/16
- Discussed formatting changes and assigned task
- Discussed changing our business question
- Assigned slides tasks to specific team members

Meeting Minutes - Tasks

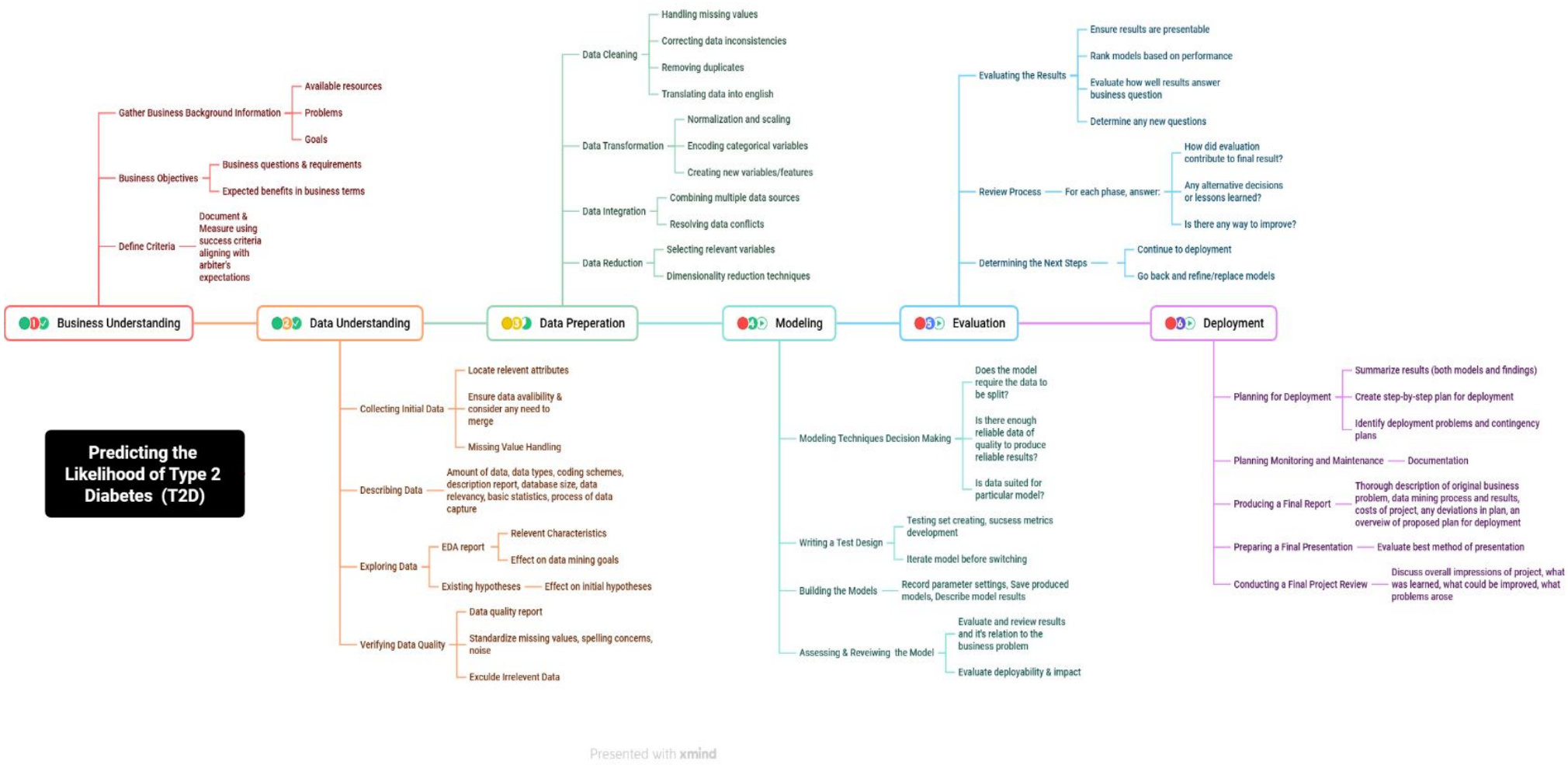
Task No.	Task Name	Description	Owner	Date	Status
T01	Collect Potential Sources	Each team member will find 1 reliable source and complete our literature review and synthesis matrix on	Tyson, Nithila, Alizeh	01/31/2026	02/05/2026
T02	Research Data Options	Find Data that tracks a unique identifier information, data that would allow for time series methods to be used	Tyson, Nithila, Alizeh	02/05/2026	02/05/2026
T03	Workflow Diagram	Creating the workflow diagram through Xmind	Nithila	02/05/2026	02/08/2026
T04	Black Box Diagram	Creating the back box diagram through Canva	Alizeh	02/05/2026	02/08/2026
T05	Class Folder Structure	Created and shared class folder	Alizeh	02/05/2026	02/08/2026
T06	Fishbone Diagram	Creating the black box diagram through Canva	Alizeh	02/05/2026	02/08/2026
T07	Business Understanding	Completing the business understanding slide	Tariq, Alizeh, Tyson, Nithila	02/05/2026	02/05/2026
T08	Data Search	Completing the data source slides	Tyson	02/05/2026	02/08/2026
T09	Analytic Problem	Completing the analytic problem slide	Tariq	02/05/2026	02/08/2026

Meeting Minutes - Tasks (cont.)

Task No.	Task Name	Description	Owner	Date	Status
T10	Review Previous Slides	Review formatting and content on previous slides based on in-class feedback	Nithila	02/10/2026	02/10/2026
T11	Project Workflow Diagram Slides	Recreate project workflow diagram to resemble a more time based flow and update it with new information	Nithila	02/12/2026	02/16/2026
T12	Code Workflow Diagram Slides + Business Understanding Slides	Create code workflow diagram slides + business understanding and analytic slides	Tariq	02/12/2026	02/16/2026
T13	Analytic Problem	Complete/revised analytic problem slides and making changes to fishbone diagram	Alizeh	02/12/2026	02/16/2026
T14	Data Understanding and Preparation Slides	Complete data understanding and preparation slides	Tyson	02/12/2026	02/16/2026
T15	Project Purpose and Questions Slides	Complete project purpose and questions slides	Alizeh	02/12/2026	02/16/2026
T16	Matching Parameters and Objectives Slides	Complete matching parameters and objectives slides	Nithila	02/12/2026	02/16/2026

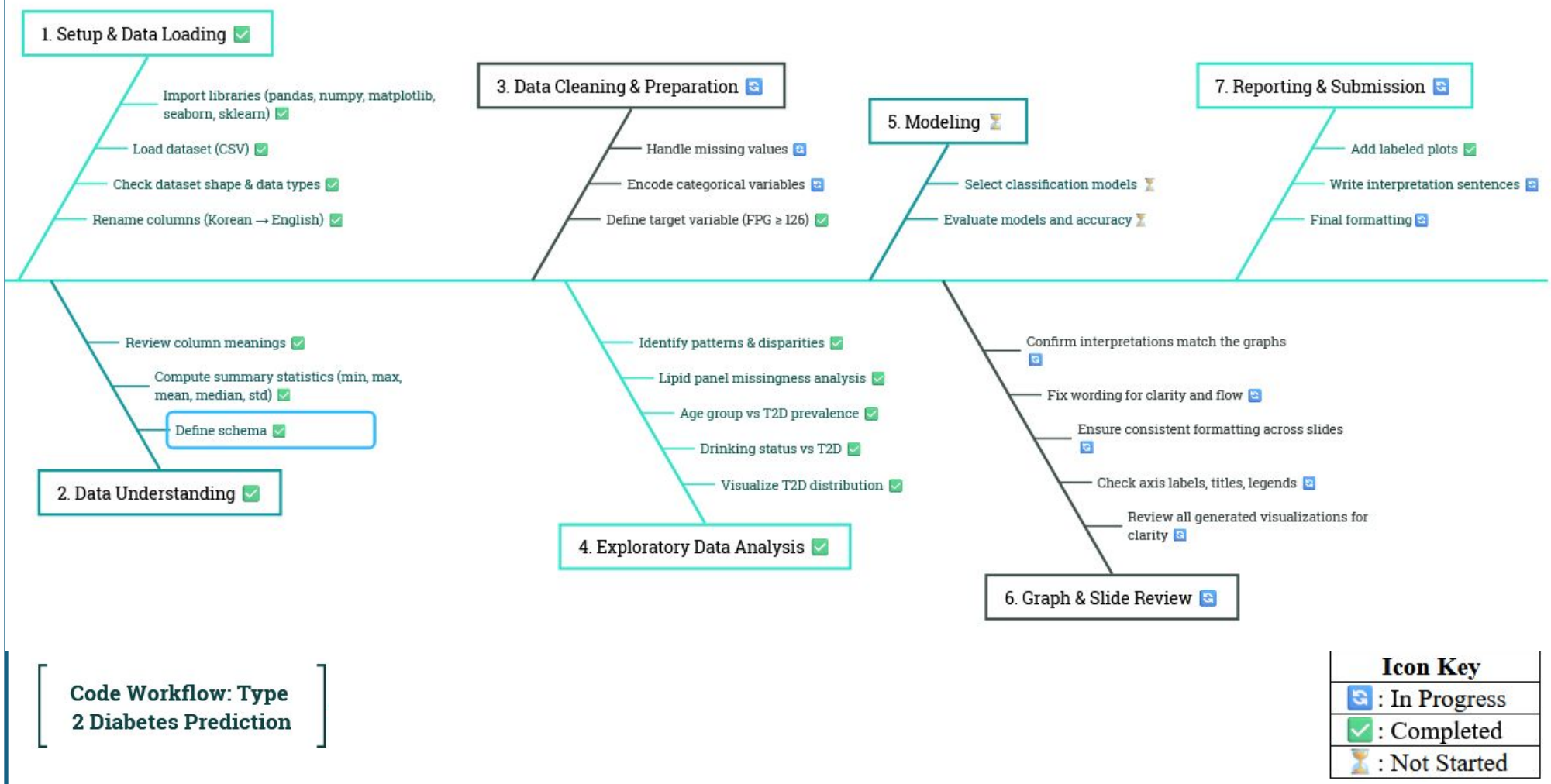
MD04 – Project Workflow Diagram

- Workflow Diagram as of 2026.02.16



MD04 – Code Workflow Diagram

- Workflow Diagram as of 2026.02.16



MD04 – Business Understanding

- Business problem and question(s)
  - Predict the likelihood that a patient has T2D using their patient data
- Problem importance
  - T2D is a significant problem that impacts the general population with serious health consequences
  - Diabetes is the 7th leading cause of death in the US (https://www.cdc.gov/nchs/fastats/diabetes.htm)
- Stakeholders
  - Patients
  - Doctors/Health care providers
  - Insurance companies
- Motivation
  - Provide early screening to identify T2D test candidates
- Suggested title for the project
  - Screening T2D Using Patient Data
- Possible challenges with the problem topic
  - Because the data is specifically of South Korean patients, the model may not be generalizable

MD04 – Analytic Problem

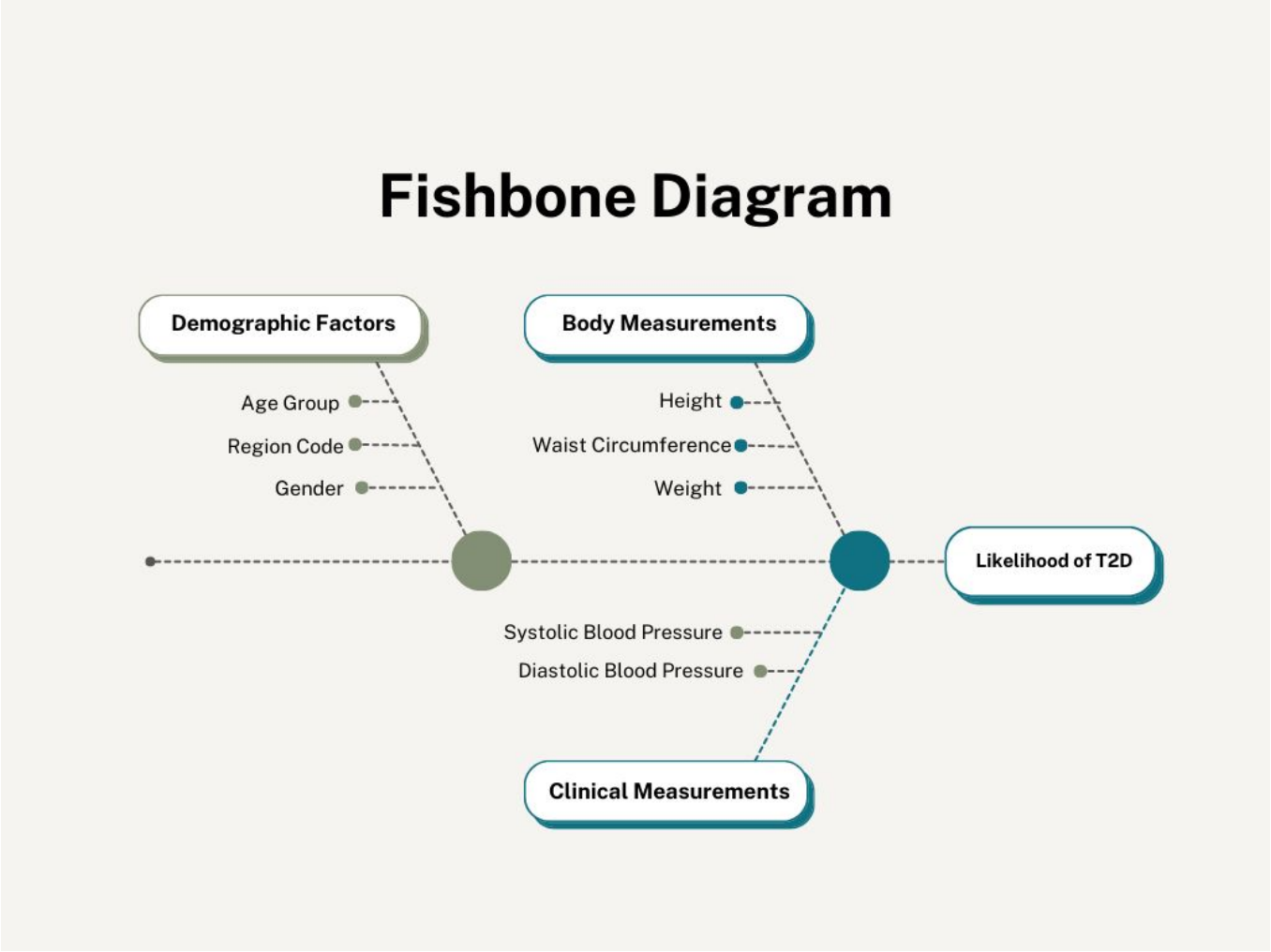
- Data:
  - 2024 health exam dataset with almost 1 million adult patients (age 20+)
  - 34 structured features including demographic, body, and clinical variables
- Model:
  - Test different models such as logistic regression, random forest, ANN, etc.
- Assumptions:
  - Data accurately reflects patient health
  - Features are sufficient to estimate risk
  - Population is reasonably representative
- Limitations:
  - Limited feature set
  - Predictions do not replace medical diagnosis

MD04 – Analytic Problem (cont.)

- Goals:
  - Accurately identify high-risk individuals
  - Develop a deployable screening tool
  - Support early intervention decisions
- Important categories and factors
  - Demographic factors
    - Age group
    - Gender
    - Region
  - Body measurements
    - Height
    - Weight
    - Waist circumference
  - Clinical measurements
    - Systolic blood pressure
    - Diastolic blood pressure
  - Blood sugar indicator
    - Fasting blood glucose

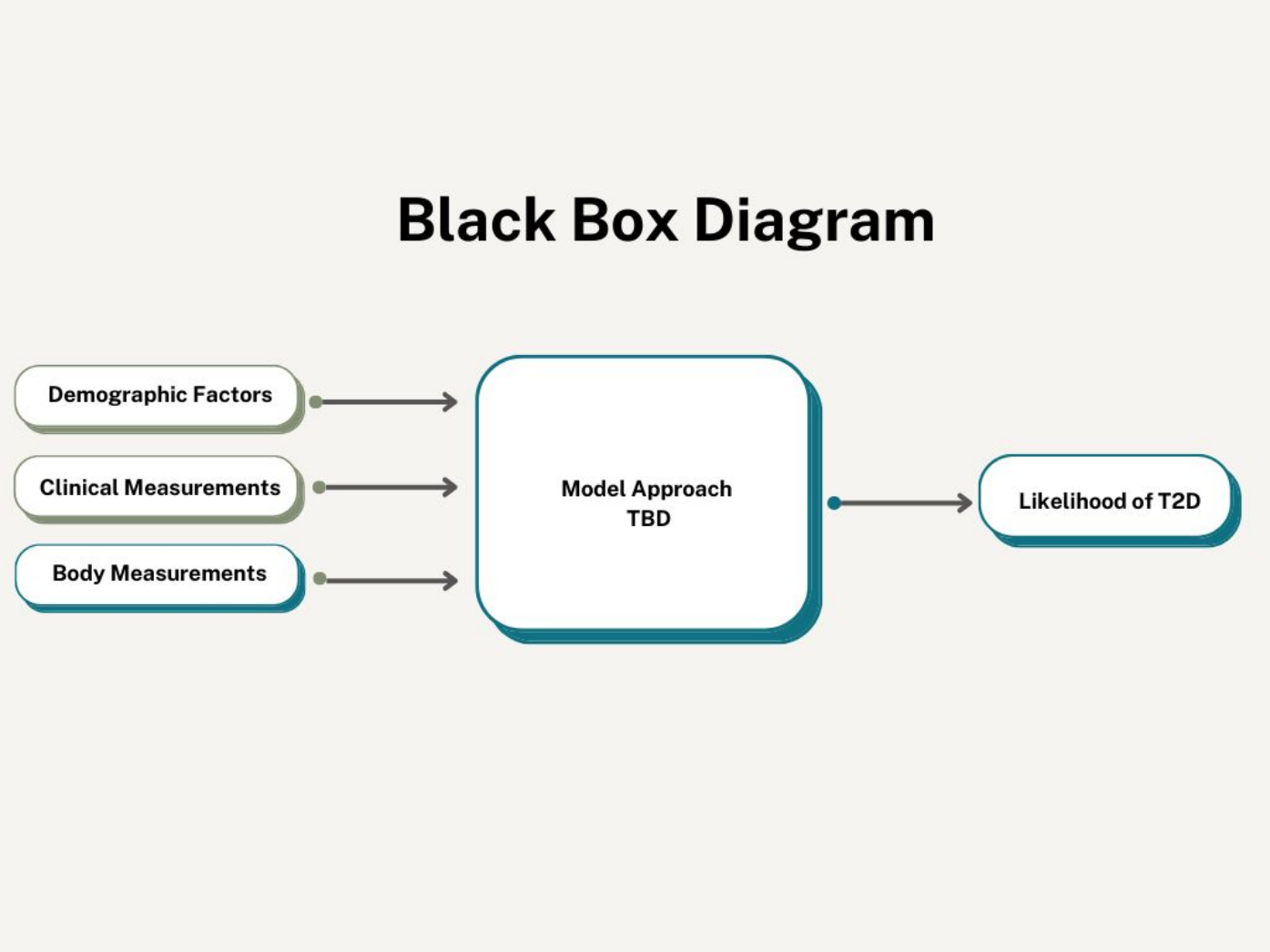
MD04 – Analytic Problem: Fishbone Diagram

- Fishbone diagram (revised)



MD04 – Analytic Problem: Black Box Diagram

- Black box diagram (revised)



MD04 – Data Search

- Data sources
  - https://www.data.go.kr/en/data/15007122/fileData.do
  - Publicly provided by South Korea's National Health Insurance Service (NHIS)
- Relevance of the data to the project
  - Patient data of 1 million randomly selected patient exams throughout year
  - Patient data from 2002-2024 (updated annually) available
  - Vast majority of exams include resting blood glucose measurements
- Possible challenges with the data
  - Some variables are rounded
  - Data and manual recorded in Korean, requiring accurate translations
  - Lack of identifying information prevents tracking patient over time

MD04 – Data Description

- Entries: 1,000,000
  - Represents an insurance-covered patient aged 20+
- Columns: 33 (27 usable features + 1 target variable)
  - Removed identifier, constant, and three unpopulated dental columns
  - Includes anthropometric details, health exam results, questionnaire answers
  - Translated from Korean to English
- File type: CSV (cp949 encoding)
- File size: 87.74 MB
- Feature types:
  - Continuous numeric: 16
  - Ordinal/binned: 3
  - Categorical/binary: 8
- Additional information:
  - Varying sparsity because not all patients received all tests
  - All data in this set is from 2024

MD04 – Column Statistics

Feature	% Missing	Minimum	Maximum	Mean	Median	St. Deviation
age_group_code*	0.0	5	18	10.67	11	3.07
height**	0.0	130	195	162.73	165	9.36
weight**	0.0	30	135	64.56	65	14.02
waist_circumference	0.05	6.8	999	81.56	81.2	10.94
systolic_blood_pressure	0.58	60	271	122.7	121	14.6
diastolic_blood_pressure	0.58	11	182	75.22	75	10.36
fasting_blood_sugar	0.58	17	991	100.98	96	23.00
total_cholesteral	65.76	25	2204	196.79	195	43.42
triglycerides	65.76	1	4622	129.32	105	99.46
hdl_cholesterol	65.76	1	908	57.09	55	15.48
ldl_cholesterol	65.76	0	2174	114.47	113	39.31
hemoglobin	0.59	0.1	25	14.22	14.2	1.53
serum_creatinine	0.59	0.06	87.2	0.84	0.8	0.37

*Grouped as 5 years (ex. 20-24). **Rounded down to nearest 5.

MD04 – Column Statistics (cont.)

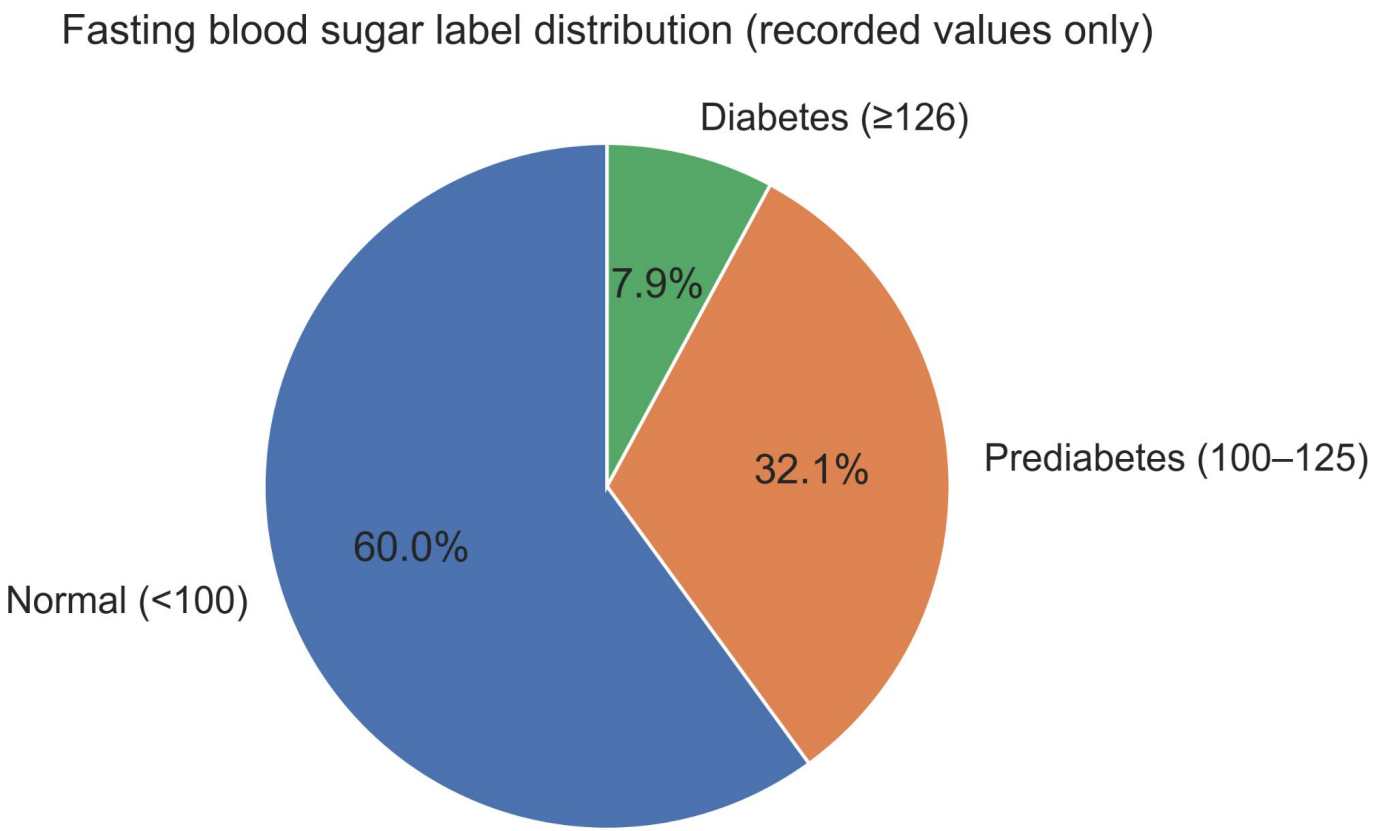
Feature	% Missing	Minimum	Maximum	Mean	Median	St. Deviation
serum_got_ast	0.58	1	7540	26.97	23	22.64
serum_gpt_alt	0.58	1	5310	26.53	21	25.52
gamma_gtp	0.58	1	9999	34.88	22	58.61
vision_left	0.02	0.1	9.9	0.95	0.9	0.58
vision_right	0.02	0.1	9.9	0.95	0.9	0.57

MD04 – Column Statistics (cont.)

Feature	% Missing	# Unique
city_code	0.0	17
gender_code	0.0	2
hearing_left	0.01	3
hearing_right	0.01	3
urine_protein	1.13	6
smoking_status	0.01	3
alcohol_consumption	0.0	2
oral_exam_status	0.0	2
dental_caries_presence	65.46	2
plaque_presence	65.46	3

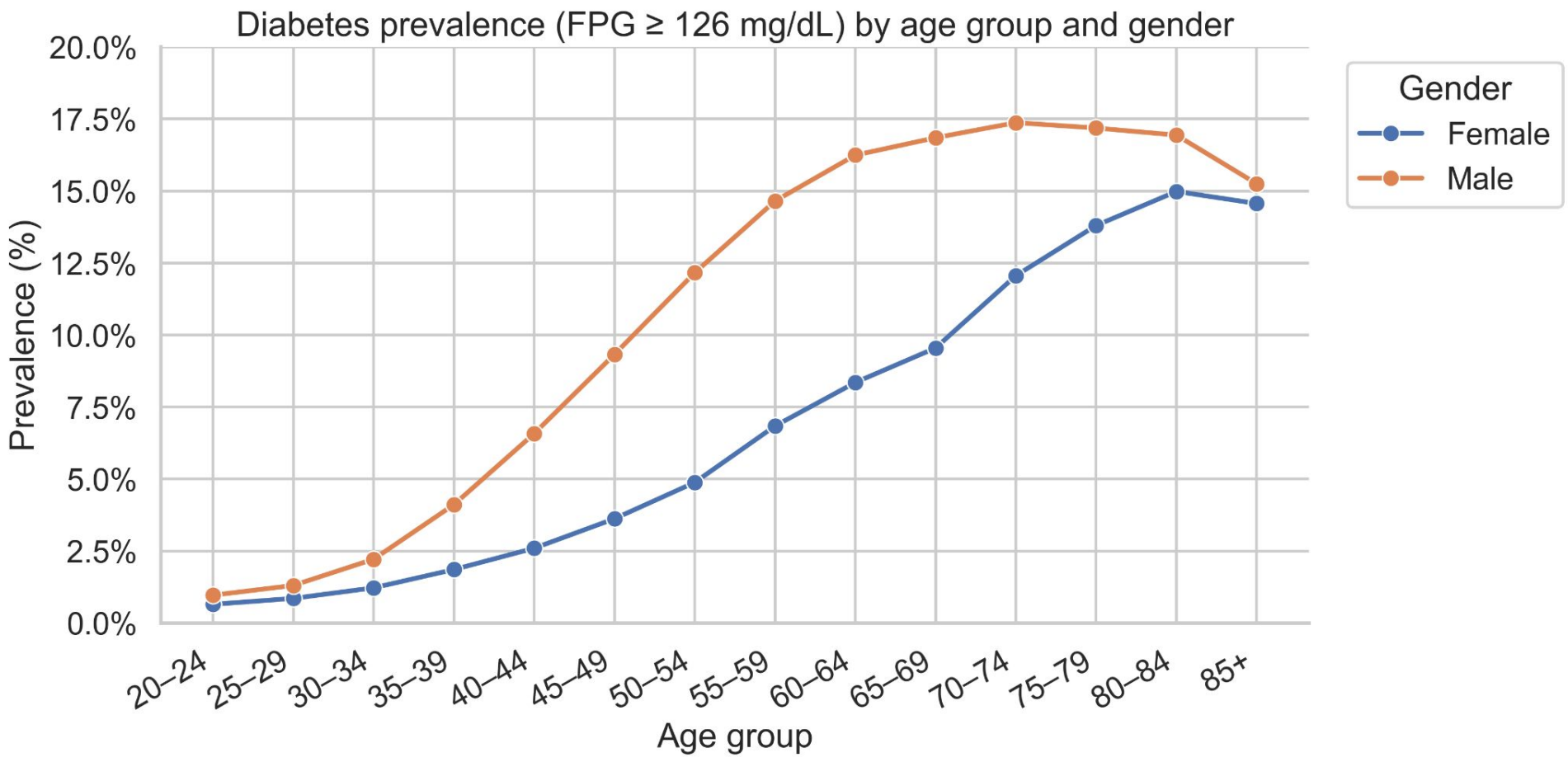
MD04 – Data Visualization

7.9% of patients can be diagnosed with T2D; 32.1% qualify as prediabetic.



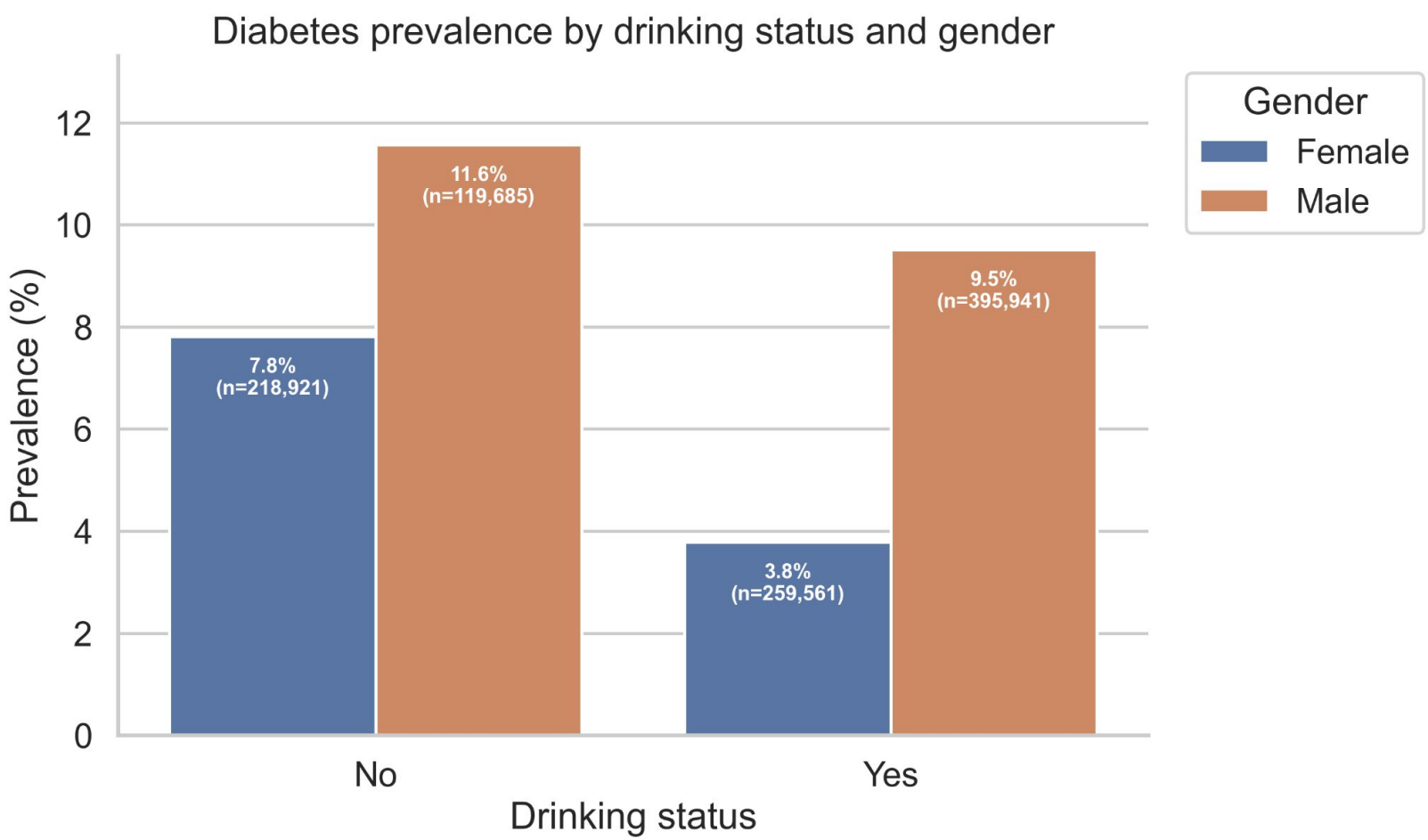
MD04 – Data Visualization (cont.)

Older age groups have higher prevalence of T2D; men have higher prevalence than women.



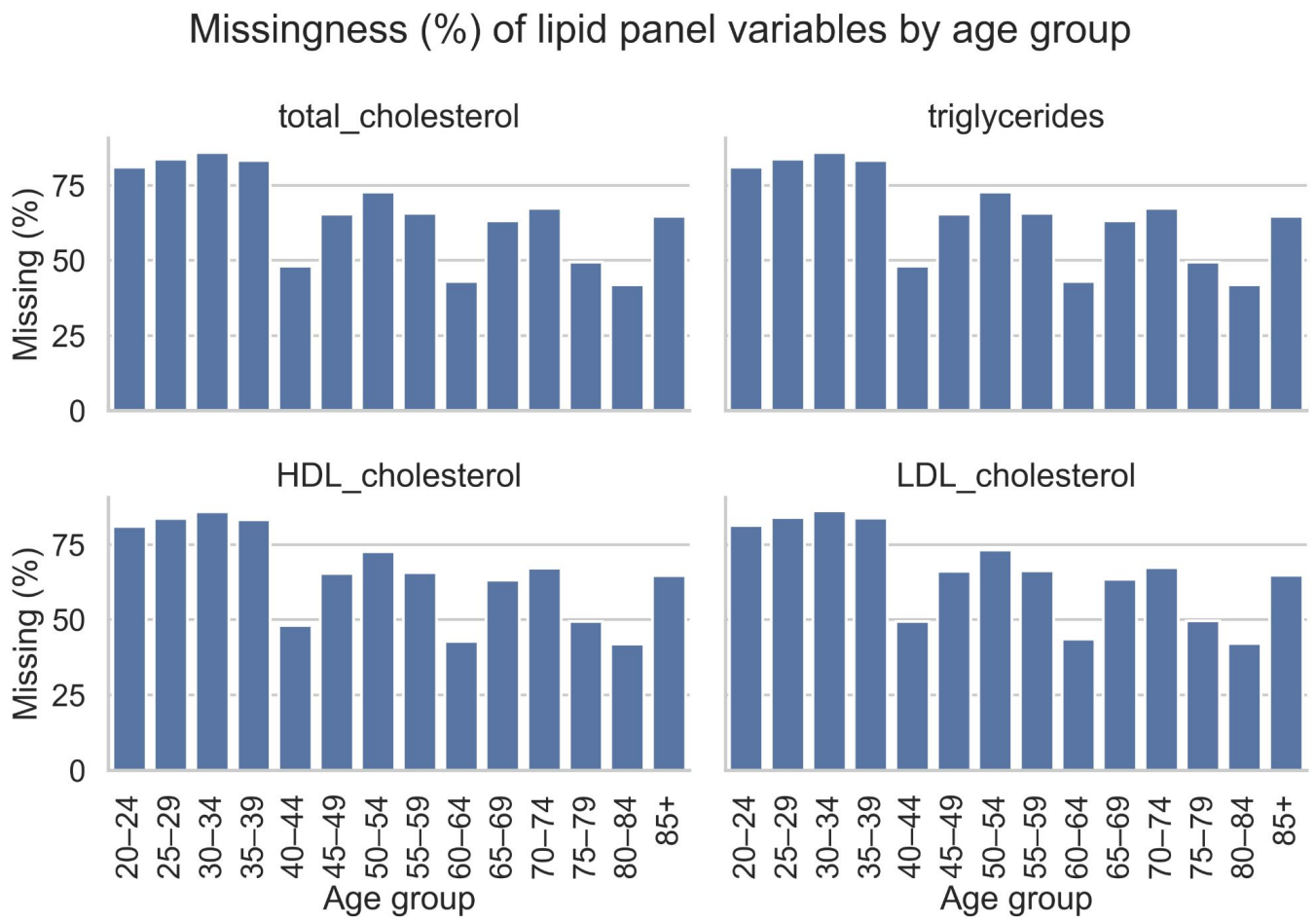
MD04 – Data Visualization (cont.)

Patients who report to not drink have higher prevalence of T2D; higher disparity observed between genders when patient reports to drink.



MD04 – Data Visualization (cont.)

Seemingly follows some sort of pattern showing when people receive lipid panel testing.



MD04 – Project Purpose and Questions

- Project purpose:
  - Enable real-world screening support using patient health data
  - Benefits of the project (benefits who and how):
    - Patients:
      - Faster identification of elevated diabetes risk
      - Early lifestyle intervention opportunities
    - Healthcare Providers:
      - Efficient prioritization of high-risk patients
    - Healthcare Systems:
      - Scalable population screening
      - Stronger preventative care strategies
- Research questions:
  - What patient health exam data can be used to estimate the likelihood of T2D?
  - What is the most effective way to deploy the T2D risk model for real-world screening use?

MD04 – Project Purpose and Questions (cont.)

- Project objectives:
  - Collect and preprocess patient health exam data
  - Perform exploratory data analysis (EDA) to identify key patterns
  - Develop and compare predictive classification models
  - Evaluate model performance using appropriate metrics
  - Select the best-performing model
  - Deploy the final model in a usable format
  - Demonstrate an end-to-end analytics workflow

MD04 – Project Objectives and Parameters

	Objective	Parameters
1	Build accurate T2D prediction model	Precision, recall, AUC
2	Identify high-risk patients early	Risk threshold, sensitivity
3	Compare classification algorithms	Model performance metrics
4	Deploy usable screening tool	Interpretability, usability

Module 05

Participants:

Tyson Johnson
Nithila Neethi Devan
Alizeh Murtaza
Tariq Abdulhak

MD05 – Revisions to work from previous module(s)

- Below is a summary of changes made to previous modules and their corresponding slide numbers:
  - Business question was update on relevant slides for consistency
  - Formatting and bullet standardization
  - Added “(cont.)” to slide titles as needed
- Class folder was updated for this week
  - Link to online class folder: CDS-492

Meeting Minutes

Date: 02/19/2026

Participants:

- Tyson Johnson
- Nithila Neethi Devan
- Alizeh Murtaza
- Tariq Abdulhak

Agenda:

- Review feedback
- Assign tasks

Notes:

- Reviewed previous submission feedback
- Discussed code sharing
- Assigned tasks
- Completed slides related to objectives and parameters

Date: 02/23/2026

Participants:

- Tyson Johnson
- Alizeh Murtaza
- Tariq Abdulhak

Agenda:

- Review tasks before submission
- Formatting and proofreading

Notes:

- Reviewed uploaded work as group
- Discussed minor formatting decisions
- Discussed issues related to noise in data from rounding
- Updated task progress
- Updated class folder

Meeting Minutes - Tasks

Task No.	Task Name	Description	Owner	Date	Status
T01	Collect Potential Sources	Each team member will find 1 reliable source and complete our literature review and synthesis matrix on	Tyson, Nithila, Alizeh	01/31/2026	02/05/2026
T02	Research Data Options	Find Data that tracks a unique identifier information, data that would allow for time series methods to be used	Tyson, Nithila, Alizeh	02/05/2026	02/05/2026
T03	Workflow Diagram	Creating the workflow diagram through Xmind	Nithila	02/05/2026	02/08/2026
T04	Black Box Diagram	Creating the back box diagram through Canva	Alizeh	02/05/2026	02/08/2026
T05	Class Folder Structure	Created and shared class folder	Alizeh	02/05/2026	02/08/2026
T06	Fishbone Diagram	Creating the black box diagram through Canva	Alizeh	02/05/2026	02/08/2026
T07	Business Understanding	Completing the business understanding slide	Tariq, Alizeh, Tyson, Nithila	02/05/2026	02/05/2026
T08	Data Search	Completing the data source slides	Tyson	02/05/2026	02/08/2026
T09	Analytic Problem	Completing the analytic problem slide	Tariq	02/05/2026	02/08/2026

Meeting Minutes - Tasks (cont.)

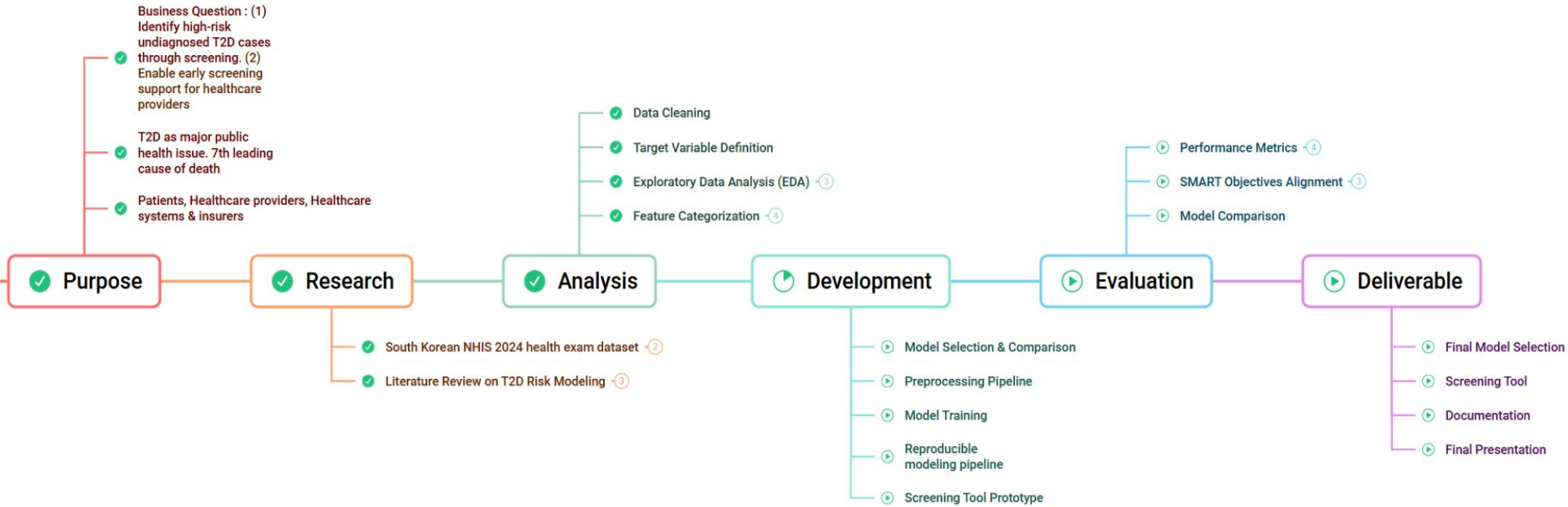
Task No.	Task Name	Description	Owner	Date	Status
T10	Review Previous Slides	Review formatting and content on previous slides based on in-class feedback	Nithila	02/10/2026	02/10/2026
T11	Project Workflow Diagram Slides	Recreate project workflow diagram to resemble a more time based flow and update it with new information	Nithila	02/12/2026	02/16/2026
T12	Code Workflow Diagram Slides + Business Understanding Slides	Create code workflow diagram slides + business understanding and analytic slides	Tariq	02/12/2026	02/16/2026
T13	Analytic Problem	Complete/revised analytic problem slides and making changes to fishbone diagram	Alizeh	02/12/2026	02/16/2026
T14	Data Understanding and Preparation Slides	Complete data understanding and preparation slides	Tyson	02/12/2026	02/16/2026
T15	Project Purpose and Questions Slides	Complete project purpose and questions slides	Alizeh	02/12/2026	02/16/2026
T16	Matching Parameters and Objectives Slides	Complete matching parameters and objectives slides	Nithila	02/12/2026	02/16/2026

Meeting Minutes - Tasks (cont.)

Task No.	Task Name	Description	Owner	Date	Status
T17	Project Workflow Diagram	Recreate/update workflow diagram structure	Nithila	02/19/2026	02/23/2026
T18	Code Workflow Diagram	Update code workflow diagram	Tariq	02/19/2026	02/23/2026
T19	Business Understanding & Project Purpose	Update business understanding & project purpose slides	Tariq	02/19/2026	02/23/2026
T20	Objectives and SMART (1)	Objectives and SMART slide completion	Nithila, Tyson, Tariq, Alizeh	02/19/2026	02/19/2026
T21	Analytic Problem (Blackbox)	Update analytic problem slides	Alizeh	02/19/2026	02/23/2026
T22	Data Preparation	Create data preparation slide based on what has been completed	Tyson	02/19/2026	02/23/2026
T23	EDA Correlations	Create correlation graphs and complete slides	Tariq	02/19/2026	02/23/2026
T24	EDA Imbalance	Complete imbalance slides	Nithila	02/19/2026	02/23/2026
T25	EDA Statistical Significance	Complete statistical significance slides	Alizeh	02/19/2026	02/23/2026
T26	Project Objectives & Parameters	Complete project objectives and parameters slide	Nithila, Tyson, Tariq, Alizeh	02/19/2026	02/19/2026
T27	Revisions	Added “(cont.)” to slide titles as needed	Tyson	02/23/2026	02/23/2026

MD05 – Project Workflow Diagram

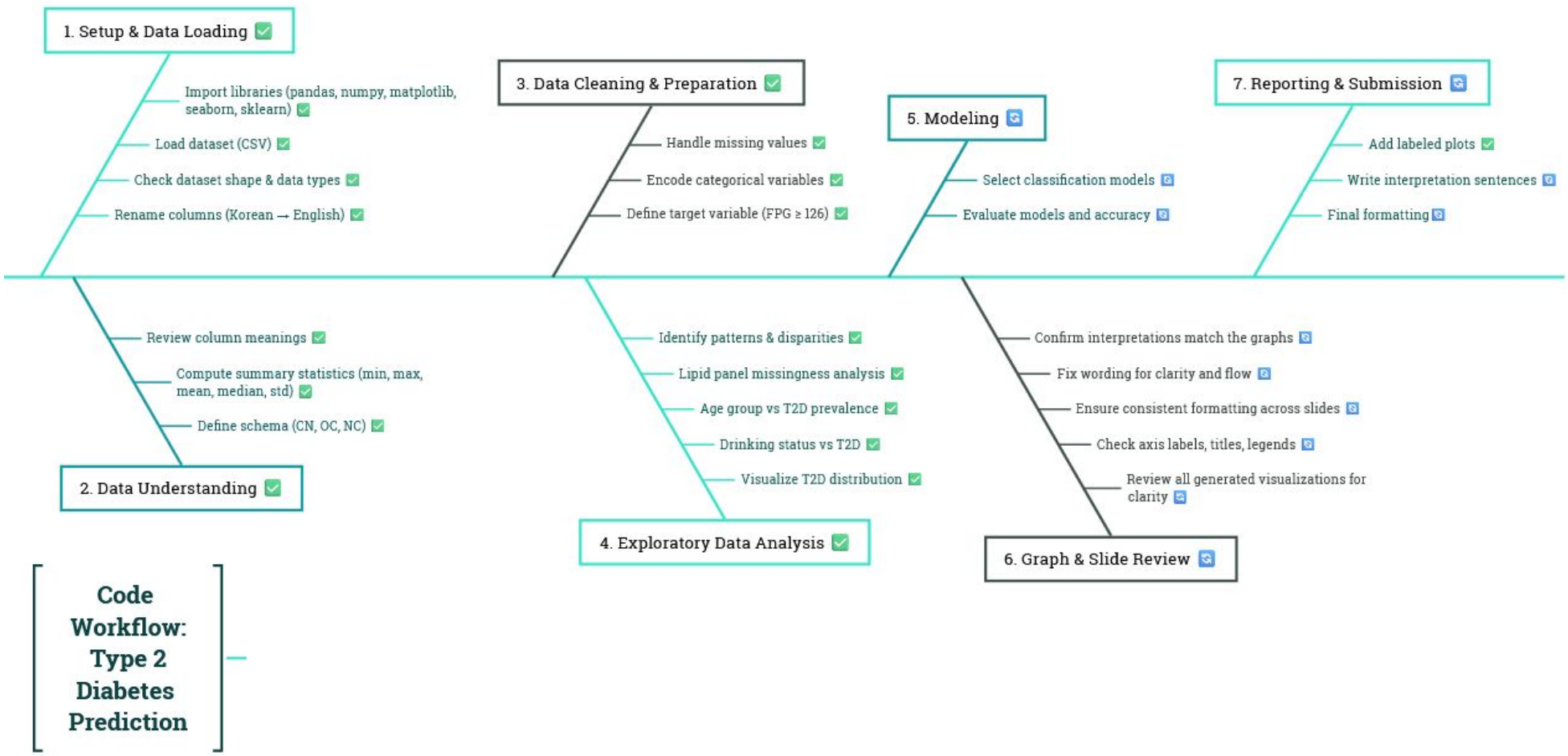
- Below is our project Workflow Diagram as of 2026.02.23



Predicting the Likelihood of Type 2 Diabetes (T2D)

MD05 – Code Workflow Diagram

- Below is our project Workflow Diagram as of 2026.02.23



MD05 – Business Understanding

- Business problem and question(s)
  - What patient health exam data can be used to estimate the likelihood of T2D?
  - What is the most effective way to deploy the T2D risk model for real-world screening use?
- Problem importance
  - T2D is a significant problem that impacts the general population with serious health consequences
  - Diabetes is the 7th leading cause of death in the US (https://www.cdc.gov/nchs/fastats/diabetes.htm)
- Stakeholders
  - Patients
  - Doctors/Health care providers
  - Insurance companies
- Motivation
  - Provide early screening to identify T2D test candidates
- Suggested title for the project
  - Predicting the Likelihood of Type 2 Diabetes (T2D)
- Possible challenges with the problem topic
  - Model may struggle with generalization due to population bias in data (South Korean)

WK05 – Objectives and SMART

Objective 1	Identify high-risk undiagnosed T2D cases through screening
Specific	Create model that can identify undiagnosed patients with high risk of having T2D
Measurable	Statistically outperforms all other tested models considered during planning
Achievable	Experience with ML-based and statistical-based modeling in supervised training
Relevant	Aligns with motivation to identify T2D test candidates
Time-bound	To be completed by the end the first week of May

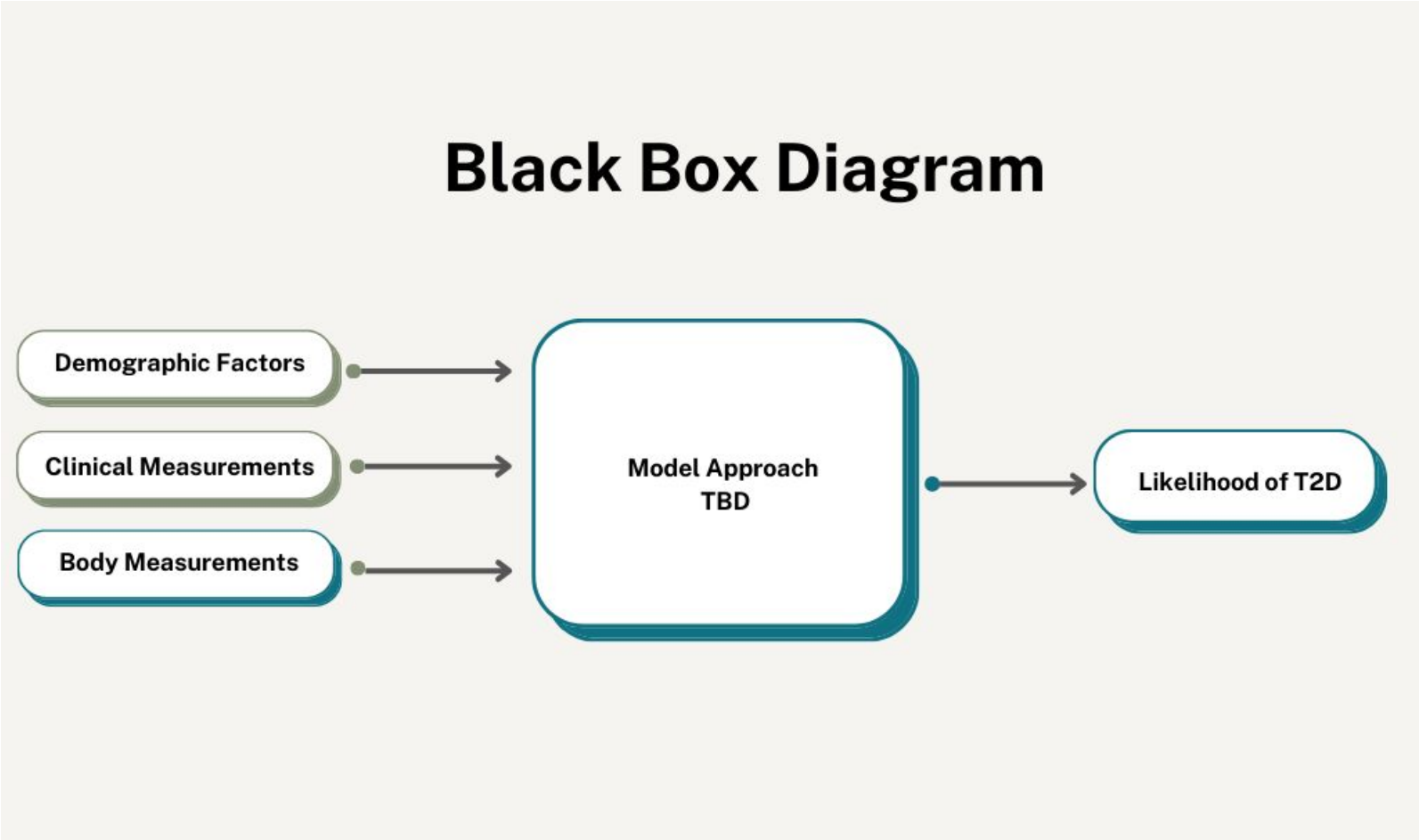
Objective 2	Release screening tool for practical application
Specific	Release tool implementing developed screening model online providing patient T2D risk score
Measurable	Accessible from any computer with internet access/publishing
Achievable	Sufficient python-based program or webpage knowledge within team
Relevant	Aligns with motivation to provide screening to identify T2D test candidates
Time-bound	To be completed by the end the first week of May

MD05 – Analytic Problem

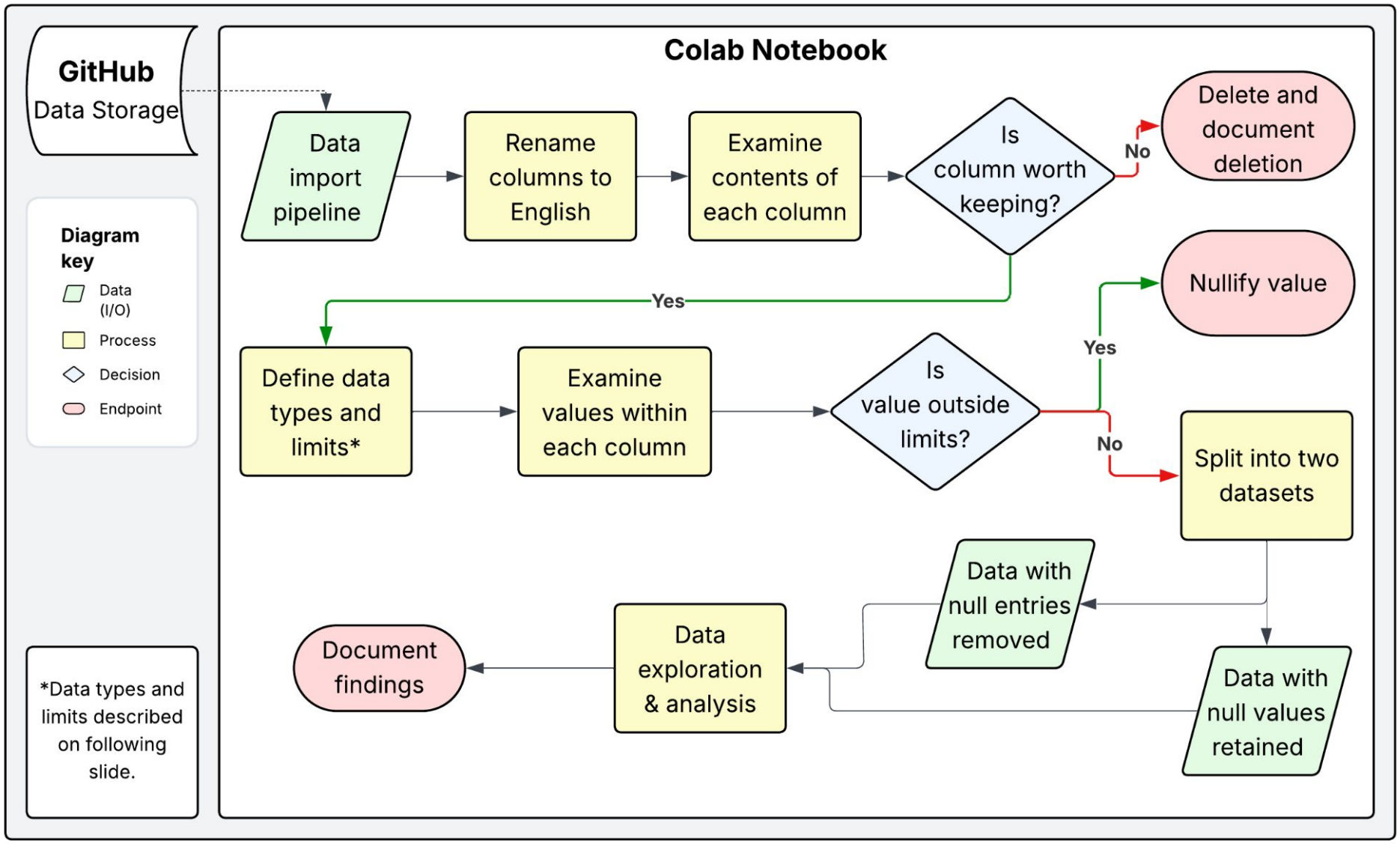
Business Objective Area	Description
Data	• 2024 health exam dataset with 1 million adult patients (age 20+) • 32 structured features including demographic, body, and clinical variables
Model	• Test different models such as logistic regression, random forest, ANN, etc.
Goals	• Accurately identify high-risk individuals • Develop a deployable screening tool
Assumptions & Limitations	• Data accurately reflects patient health • Features are sufficient to estimate risk • Population is reasonably representative • Limited feature set • Predictions do not replace medical diagnosis

MD05 – Analytic Problem: Black Box Diagram

- Black box diagram (revised)



MD05 – Data Cleaning



MD05 – Data Schema

Column Name	Data Type	Domain	Details
sex_code	int (NC)	{1, 2}	1=female, 2=male
city_code	int (NC)	{11, 26-31, 36, 41-49}	unique codes for cities/regions
hearing_left	int (NC)	{1, 2}	1=normal, 2=abnormal
hearing_right	int (NC)	{1, 2}	1=normal, 2=abnormal
alcohol_consumption	int (NC)	{0, 1}	0=does not drink, 1=does drink
oral_exam	int (NC)	{0, 1}	0=not received, 1=received
caries_presence	int (NC)	{0, 1}	0=absent, 1=present
plaque_presence	int (NC)	{0, 1}	0=absent, 1=present
age_group_code	int (OC)	{5-18}	5: [20-24], …, 18: 85+
smoking_status	int (OC)	{1, 2, 3}	1=never, 2=former, 3=current
urine_protein	int (OC)	{1-6}	1: (−), 2: (±). 3: (+1), ..., 6: (+4)
height	int (CN)	[100, 230]	round down to nearest 5cm
weight	int (CN)	[20, 250]	round down to nearest 5kg
waist_circumference	float (CN)	[30, 250]	round to nearest 0.1cm

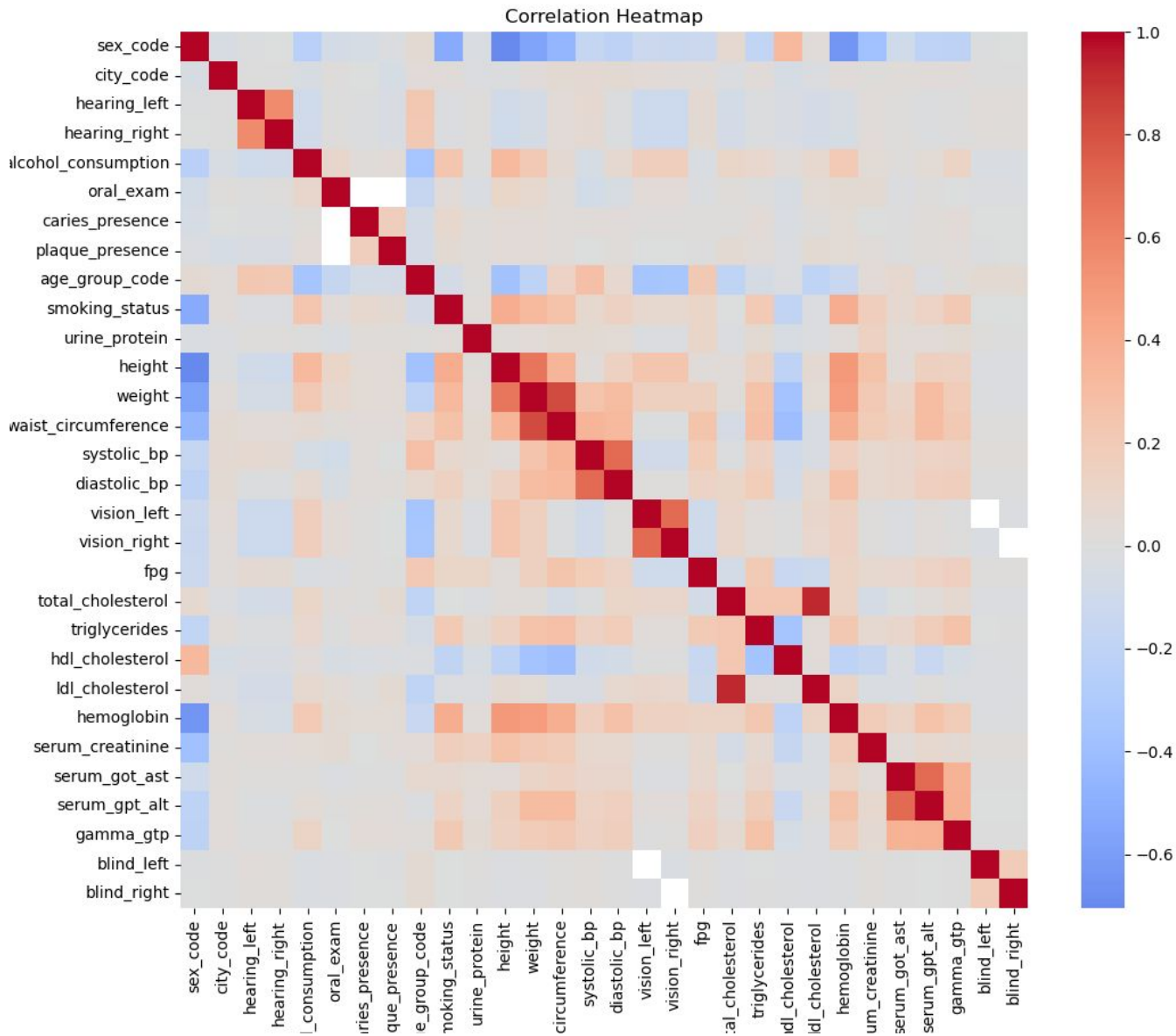
NC: Nominal Category
OC: Ordinal Category
CN: Continuous Numeric

MD05 – Data Schema (cont.)

Column Name	Data Type	Domain	Details
systolic_bp	int (CN)	[60, 300]	mmHg
diastolic_bp	int (CN)	[30, 200]	mmHg
vision_left	float (CN)	[0.1, 2.5]	9.9 for blindness
vision_right	float (CN)	[0.1, 2.5]	9.9 for blindness
fpg	int (CN)	[30, 1000]	mg/dL
total_cholesterol	int (CN)	[50, 1000]	mg/dL
triglycerides	int (CN)	[10, 5000]	mg/dL
hdl_cholesterol	int (CN)	[5, 200]	mg/dL
ldl_cholesterol	int (CN)	[5, 700]	mg/dL
hemoglobin	float (CN)	[3, 25]	g/dL
serum_creatinine	float (CN)	[0.1, 30]	mg/dL
serum_got_ast	int (CN)	[1, 5000]	U/L
serum_gpt_alt	int (CN)	[1, 5000]	U/L
gamma_gtp	int (CN)	[1, 5000]	U/L

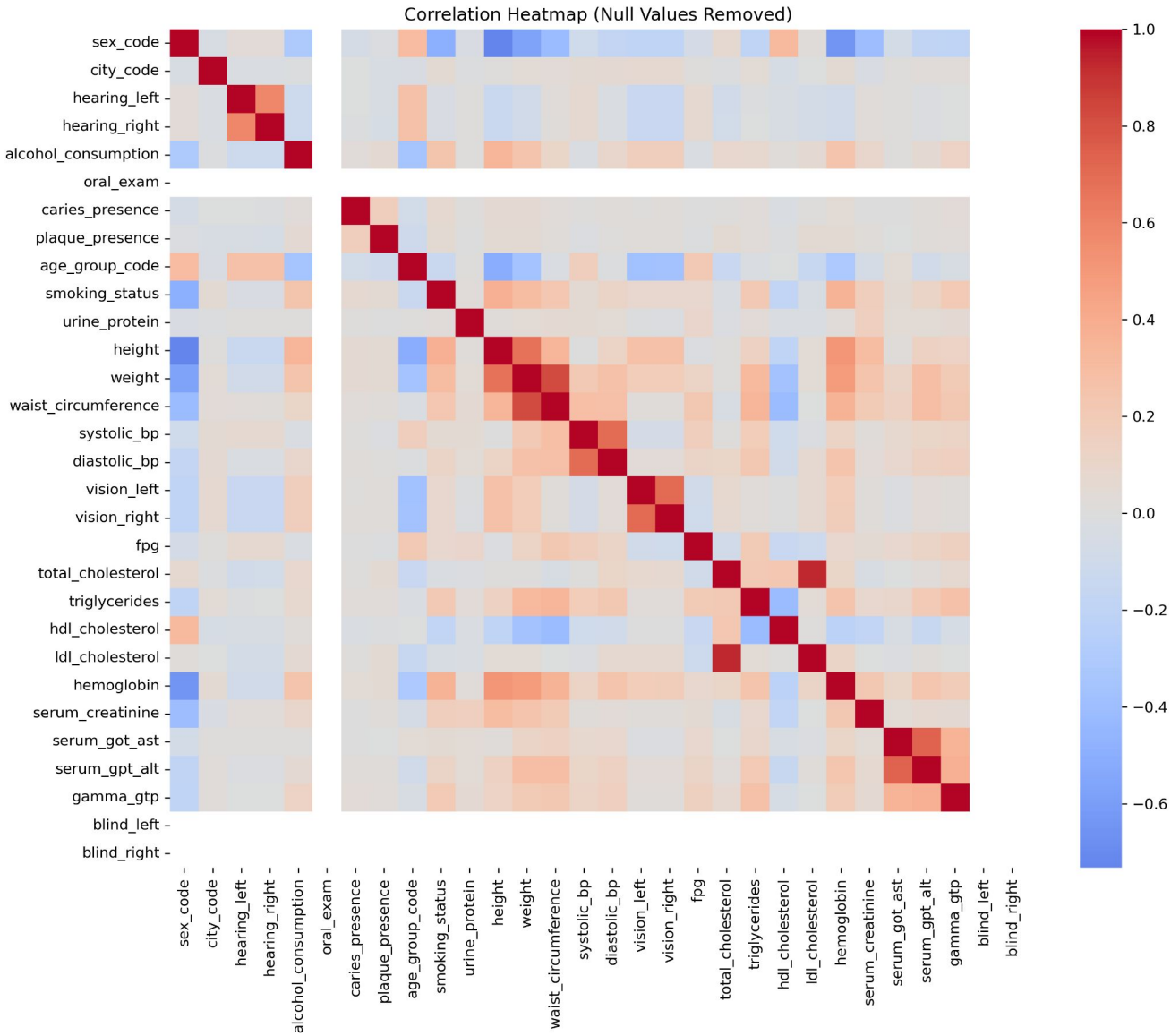
NC: Nominal Category
OC: Ordinal Category
CN: Continuous Numeral

MD05 – Data Exploration: Correlations



- Fasting Plasma Glucose (FPG) has a strong correlation with:
  - Waist circumference
  - Triglycerides
  - Blood pressure
- HDL cholesterol has a negative correlation with triglycerides
- Weight and waist circumference are strongly correlated

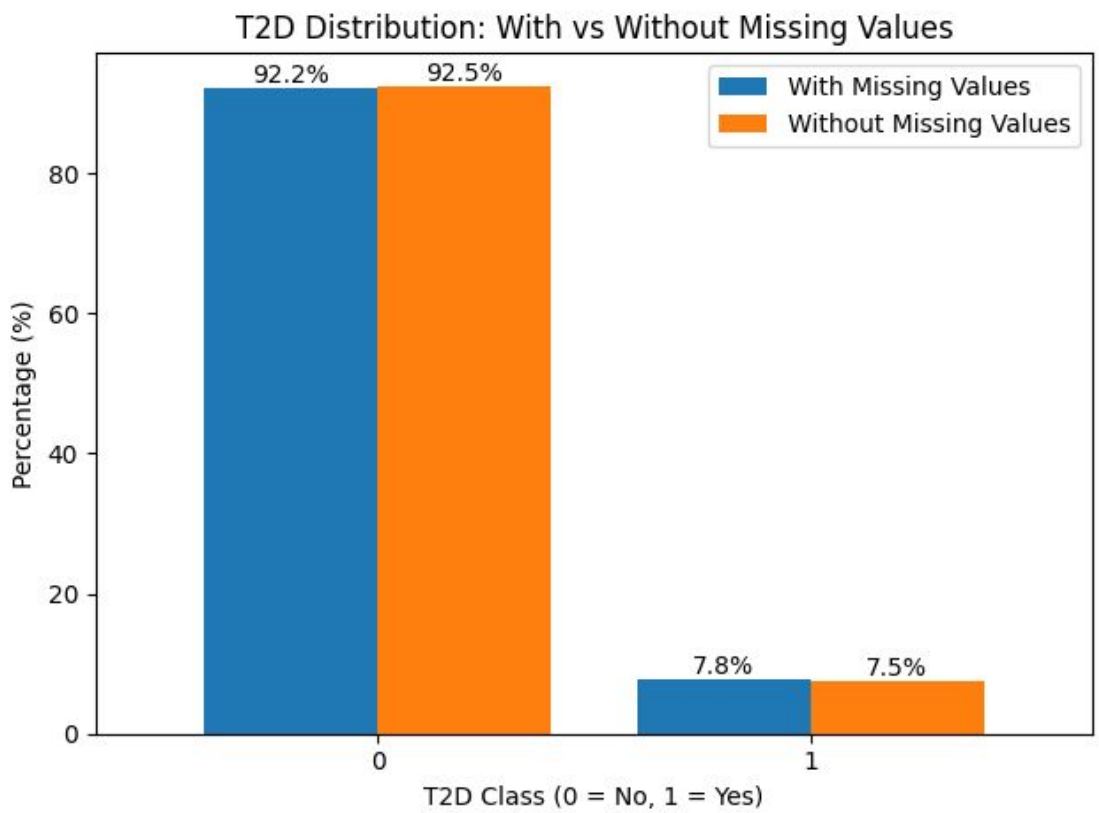
MD05 – Data Exploration: Correlations (cont.)



- Strong positive correlation between weight and waist circumference
- Clear clustering with glucose, lipids, and blood pressure
  - Shows interconnected cardiometabolic patterns
- Fasting Plasma Glucose (FPG) aligns positively with obesity and lipid markers

MD05 – Data Exploration: Imbalance

- Class Distribution:
  - **Non-T2D**: 921,528 patients **(92.15%)**
  - **T2D** : 78,472 patients **(7.85%)**
  - **Imbalance Ratio: 11.74 : 1** (Majority : Minority)
- Modeling Implications:
  - Strong class imbalance detected
  - Accuracy alone is misleading, recall and precision are critical for evaluation
  - Class distribution is consistent before and after null removal
    - Cleaning did not alter the imbalance much



MD05 – Data Exploration: Statistical Significance of Difference

Statistical Test	Type of Test	df_clean Results	df_final Results	Conclusion
Test 1: Gender vs T2D	Chi-square test of independence	• $\chi^2(1) = 6578.53$ • $p < 0.001$ • Cramer's V = 0.081	• $\chi^2(1) = 908.05$ • $p < 0.001$ • Cramer's V = 0.055	Significant association between gender and T2D in both datasets. Effect size is small but consistent.
Test 2: Waist Circumference vs T2D	Independent t-test	• $t = 176.47$ • $p < 0.001$ • Mean diff = 6.73 cm	• $t = 84.77$ • $p < 0.001$ • Mean diff = 5.38 cm	T2D patients have significantly higher waist circumference. Clinically meaningful difference persists.
Test 3: Age Group vs T2D	Chi-square test of independence	• $\chi^2(13) = 29926.14$ • $p < 0.001$	• $\chi^2(13) = 7439.34$ • $p < 0.001$	T2D prevalence differs significantly across age groups. Strong and stable association.
Test 4: Waist-Weight Index (WWI) vs T2D	Independent t-test	• $t = 25.22$ • $p < 0.001$ • Mean diff = 0.0168	• $t = 18.94$ • $p < 0.001$ • Mean diff = 0.0217	T2D prevalence differs significantly across age groups. Strong and stable association.

MD05 – Project Objectives and Parameters

	Objective	Parameters
1	Identify high-risk undiagnosed T2D cases through screening	Number of at-risk patients correctly flagged/reduce in missed cases
2	Release screening tool for practical application	Focus on public accessibility Important considerations: • Cost • Production time